

Elastoplastic semi-analytical investigation on a deep circular tunnel incorporating generalized Zhang–Zhu rock mass strength

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ABSTRACT

The semi-analytical method based on finite difference method is widely used in the study of the mechanical response of surrounding rocks in a circular tunnel. The core of the three-dimensional (3D) mechanical analysis of the plastic zone is to solve nonlinear equations with a complex yield criterion, which leads to a long iterative solution time and low accuracy. In this study, a new 3D semi-analytical method is proposed for a circular tunnel excavation. The method is based on the finite difference method, wherein a linear equation is used to replace the complex yield surface equation. This significantly reduces the iterative solution process and effectively improves the calculation rate. The accuracy of the proposed method is verified using conventional and 3D numerical methods. The two-dimensional (2D) generalized Hoek–Brown criterion (GHB), 3D generalized Pan–Hudson criterion (GPH), and 3D generalized Zhang–Zhu criterion (GZZ) are used to analyze the applicability of the proposed method. The mechanical properties of a circular tunnel excavation are studied using these criteria. Our results show that the calculation rate of the plastic zone response of the proposed method (based on the GZZ criterion) is 12,000 times higher than that of the conventional iterative method. The overall calculation rate is increased by a factor of 65, and the calculation error is smaller than that of the conventional method. The GHB does not consider the intermediate principal stress, and the GPH cannot reflect the mechanical properties of rock masses with different tensile and compressive strengths, although it considers the intermediate principal stress. The convergence–constraint characteristics of the surrounding rock of the GHB and GPH are significantly different from those of the GZZ. There is an optimal initial axial stress value q_{opt} that yields the smallest deformation of the surrounding rock and the best stability. The internal support pressure and residual strength of the rock mass have a significant effect on the mechanical response of the surrounding rock. The proposed method has acceptable applicability to elastic–brittle and elastic–perfectly plastic rock masses. Additionally, it can be used in the study of deep-buried tunnel excavation at high geo-stresses.

1. Introduction

The mechanical response of the surrounding rock after tunnel excavation is a complicated problem. Large-scale numerical calculations are often required to achieve fine design in practical engineering. Analytical or semi-analytical methods are preferred in parameter research, constitutive model verification, and analysis of nonlinear mechanics because of their simplicity and fast calculation speed, as compared with the complex modeling analysis of commercial software. The elastic–plastic mechanical response analysis of a circular tunnel excavation is a classic rock mass mechanics problem, which is useful in studying the convergence–constraint characteristics of the surrounding

rock in depth. Moreover, it is important for rapid preliminary design.

In terms of the description of the mechanical behavior of the surrounding rock, Hoek and Brown (1997) noted that the elastic–brittle model is suitable for high-quality rock mass, elastic–perfectly plastic model is suitable for poor quality rock mass, and strain–softening model is suitable for average quality rock mass. Although the strain–softening model can be applied to a wider range of rock masses, the softening models used by different researchers in the analysis were not the same (Ghorbani and Hasanzadehshooili, 2019; Han et al., 2013; Wang and Qian, 2018). Moreover, the results obtained using different softening laws were significantly different. In addition, hard rock exhibits mainly brittle failure at a low-stress state, whereas it may exhibit ductile failure

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at a high-stress states (Feng et al., 2021; Xie et al., 2018). Therefore, the elastic–brittle model and elastic–perfectly plastic model can be used to describe the mechanical behavior of hard rock at different stress states. The elastic–brittle model (Guan et al., 2020; Wang and Yin, 2011; Wang et al., 2012; Zareifard and Fahimifar, 2016; Zhang et al., 2012a) and elastic–perfectly plastic model (Carranza-Torres and Fairhurst, 1999; Kabwe, 2020; Park, 2015; Vrakas and Anagnostou, 2014; Xu and Yu, 2006) are widely used in the response analysis of the surrounding rock. They can be regarded as two extreme cases of the strain–softening model. Their calculation results define the upper and lower limits of the mechanical response of the surrounding rock, which has very important theoretical significance.

Regarding the yield criterion, the two-dimensional (2D) linear Mohr–Coulomb criterion has high applicability because of its simple form and ease of parameter measurement. Numerous researches have used this criterion to provide elastic–plastic solutions (Alejano et al., 2009; González-Cao et al., 2013; Guan et al., 2007; Lu et al., 2010; Zareifard and Fahimifar, 2015). Subsequently, the 2D nonlinear Hoek–Brown criterion obtained from engineering experience and experimental data was found to be more suitable for engineering practice. It has been widely used in rock mass engineering (Carranza-Torres and Fairhurst, 1999; Sharan, 2003, 2005) with far-reaching effects. However, the plane strain problem using 2D strength criterion considers only the response of the surrounding rock in the plane and neglects the effect of the axial stress. In the plastic zone, the axial stress is generally the intermediate principal stress (Guan et al., 2018; Reed, 1988; Vrakas and Anagnostou, 2014). This axial stress has a significant effect on the strength and stability of the surrounding rock, which has been verified in several studies (Deng et al., 2020; Feng et al., 2019; Kaunda, 2014; Pan et al., 2012; Zhao et al., 2021). Several three-dimensional (3D) Hoek–Brown criteria have been proposed to analyze the strength characteristics of rock masses at different stress states more precisely (Jiang and Zhao, 2014; Li et al., 2021; Pan and Hudson, 1988; Priest, 2005; Zhang and Zhu, 2007) by considering the intermediate principal stress. Among these, the generalized Zhang–Zhu criterion (GZZ) proposed by Zhang and Zhu (Zhang and Zhu, 2007; Zhang, 2008) has the advantages of the Hoek–Brown strength criterion. Additionally, it has a high fitting accuracy for the rock true triaxial test data, which is one of the criteria recommended by the International Society for Rock Mechanics and Rock Engineering (ISRM) (Ulusay, 2015).

In terms of the Mohr–Coulomb criterion and the original Hoek–Brown criterion with simple expressions, analytical solutions can be obtained, whereas analytical solutions cannot be obtained for the generalized Hoek–Brown criterion (GHB) and 3D Hoek–Brown criterion (e.g., GZZ), whose expressions are transcendental equations, and semi-analytical solutions require the use of numerical methods to solve nonlinear equations and second-order differential equations (Carranza-Torres, 2004), or numerical integration calculations (Sharan, 2008). Lee and Pietruszczak (2008) proposed a semi-analytical method to solve the stress and strain components of the plastic zone by dividing the plastic zone into a finite number of concentric rings, and using the finite difference method. This method assumes that the radial stress difference of each ring is equal. Therefore, researches have proposed different solutions for the surrounding rock response using different ring division methods (Wang et al., 2010; Zhang et al., 2012b; Zou et al., 2017a), or different strength criteria, such as the Mohr–Coulomb criterion (Cui et al., 2015), unified strength theory (Ghorbani and Hasanzadehshooili, 2019), GHB (Zareifard, 2020; Zou et al., 2017b; Zou et al., 2019; Zou et al., 2020), and GZZ strength criterion (Chen et al., 2021; Xia et al., 2018; Xu et al., 2021). The semi-analytical method using concentric rings division and finite differences is widely used in the study of the nonlinear mechanical response of surrounding rock after the excavation of a circular tunnel.

The number of equations that need to be solved is higher for 3D analysis than for 2D analytical calculations. The expression of the 3D strength criterion is more complex, which significantly increases the

complexity of the calculation. The finite difference method based on the 3D Hoek–Brown criterion uses numerical methods to iteratively calculate the complex yield surface equation at each ring. Repeated iterative calculations are time consuming when calculating the empirical model curve of convergence–confinement method (CCM). In this study, the solution process was improved by avoiding the iterative solution of the stress at each point in the plastic zone. A high-speed and high-precision calculation method was proposed. The accuracy of the proposed method was verified using calculation cases. The differences in the calculation results of different strength criteria were compared. Finally, the applicability of the proposed method to the strain–softening model was analyzed.

2. Calculation model

The tunnel excavation problem can be simplified to the “thick-walled cylinder” problem, as shown in Fig. 1. A uniform hydrostatic pressure p acts around the homogeneous and isotropic surrounding rock. The axial stress q acts perpendicular to the excavation plane. After excavation, the excavation radius of the tunnel is R_0 , and the support pressure p_s acts on the wall of the tunnel. The surrounding rock stress state is related to the support pressure. The critical stress p_{cr} is defined to determine the existence of the plastic zone. When the support pressure is sufficient ($p_s > p_{cr}$), the entire surrounding rock is in an elastic state, and no plastic zone is generated. When the supporting pressure is insufficient ($p_s < p_{cr}$), a plastic zone is produced near the wall. When a plastic zone exists, p_{cr} is the radial stress at the interface of the elastic–plastic zone and R_p is the radius of the plastic zone. The excavation problem requires the solutions of the circumferential σ_θ , radial σ_r , axial stresses σ_z , and radial displacement u_r of the calculation unit at any radius r .

2.1. Rock mass strength criterion

When the excavation unloading exceeds a certain value, the plastic zone exists in the surrounding rock. The stress in the plastic zone should satisfy the consistency condition, as follows.

$$F = F(\sigma) = 0, \quad (1)$$

where $\sigma = [\sigma_r, \sigma_\theta, \sigma_z]$ is the stress vector of the rock mass. Note that F has different expressions for different strength criteria.

In 1980, Hoek and Brown (1980) proposed the influential Hoek–Brown strength criterion based on a significant number of rock triaxial tests and field rock mass test data. Considering its application in rock

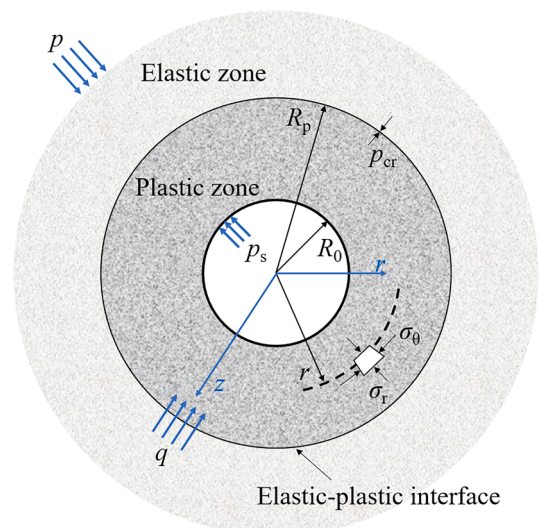


Fig. 1. Tunnel excavation model.

mass engineering, the modified GHB criterion (Hoek et al., 1992) can be expressed as:

$$F_{\text{GHB}} = \frac{1}{\sigma_c^{1/a-1}} (\sigma_1 - \sigma_3)^{1/a} - (m_b \sigma_3 + s \sigma_c) = 0, \quad (2)$$

where σ_1 and σ_3 are the maximum and minimum principal stresses, respectively; σ_c is the uniaxial compressive strength of the undisturbed rock; and m_b , s , and a are empirical parameters that reflect the characteristics of the rock mass, which can be obtained using the rock mass GSI , m_i , and D . Their transformation relationship is:

$$\begin{cases} m_b = m_i \exp[(GSI - 100)/(28 - 14D)], \\ s = \exp[(GSI - 100)/(9 - 3D)], \\ a = 1/2 + [\exp(-GSI/15) - \exp(-20/3)]/6, \end{cases} \quad (3)$$

where GSI is the geological strength index; m_i is the laboratory rock parameter, which can be directly selected based on the approach proposed by Hoek et al. (2002); and D is the disturbance parameter. Currently, the Hoek–Brown criterion parameters have formed a complete and reliable system (Hoek and Brown, 2019).

The projection of the GHB on the π -plane is a hexagon with unequal diagonals, as shown in Fig. 2. The corners of its yield surface need to be modified in the finite element calculation. More importantly, the 2D GHB does not reflect the effect of the intermediate principal stress on the rock mass, which significantly limits the 3D fine analysis of the rock mass. Numerous researches have proposed several 3D Hoek–Brown criteria considering the intermediate principal stress. Among them, the generalized Pan–Hudson criterion (GPH) and GZZ are the strength criteria recommended by the ISRM.

The Pan–Hudson criterion, a 3D Hoek–Brown criterion, was proposed by Pan and Hudson (1988). The generalized form of this criterion is expressed as:

$$F_{\text{GPH}} = \frac{1}{\sigma_c^{1/a-1}} \left(\frac{3}{\sqrt{2}} \tau_{\text{oct}} \right)^{1/a} + \frac{3}{2\sqrt{2}} m_b \tau_{\text{oct}} - m_b \frac{I_1}{3} - s \sigma_c = 0, \quad (4)$$

where τ_{oct} is the octahedral shear stress; and I_1 is the first invariant of the stress tensor, which can be expressed as:

$$\begin{cases} \tau_{\text{oct}} = \frac{1}{3} \sqrt{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2}, \\ I_1 = \sigma_1 + \sigma_2 + \sigma_3. \end{cases} \quad (5)$$

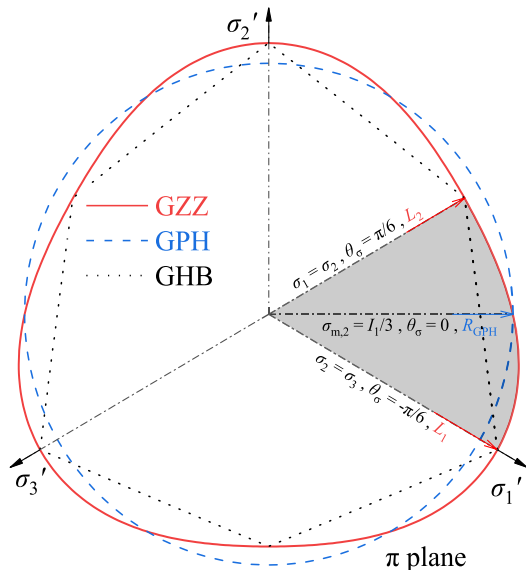


Fig. 2. Strength envelopes of different criteria on the π -plane.

As shown in Fig. 2, the projection of the GPH on the π -plane is a standard circle, which does not reflect the different tensile and compressive strengths of geotechnical materials, similar to the Drucker–Prager criterion. Therefore, GPH is only suitable for weak rock masses (Melkounian et al., 2008; Zhang, 2008). The GPH cannot be simplified to the 2D Hoek–Brown criterion at neither triaxial compression nor triaxial tension conditions (Li et al., 2021). Additionally, it cannot directly use the Hoek–Brown parameter system.

The Zhang–Zhu criterion, another 3D Hoek–Brown criterion, was proposed by Zhang and Zhu (2007), and its generalized form (i.e., GZZ) was proposed by Zhang (2008). Cai et al. (Cai et al., 2021; Zhang et al., 2013; Zhu et al., 2017) subsequently modified the GZZ problem where the criterion was not smooth on the π -plane, and embedded it in the 3D finite element calculation software GeoFBA^{3D}. A schematic of the yield surface of the GZZ in the stress space is shown in Fig. 3. The modified GZZ expressed in terms of the stress invariant (Cai et al., 2021) is:

$$F_{\text{GZZ}} = \frac{1}{\sigma_c^{1/a-1}} \left(\sqrt{3} J_2 \right)^{1/a} + \left(\frac{3 + \sin 3\theta_\sigma}{2\sqrt{3}} \right) m_b \sqrt{J_2} - m_b \frac{I_1}{3} - s \sigma_c = 0, \quad (6)$$

where θ_σ is the Lode angle; and J_2 and J_3 are the second and third invariants of the stress deviation, respectively, which are expressed as:

$$\begin{cases} \sin 3\theta_\sigma = \frac{3\sqrt{3}}{2} \frac{J_3}{J_2^{3/2}}, \\ J_2 = \frac{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2}{6}, \\ J_3 = S_1 S_2 S_3 = \left(\sigma_1 - \frac{I_1}{3} \right) \left(\sigma_2 - \frac{I_1}{3} \right) \left(\sigma_3 - \frac{I_1}{3} \right). \end{cases} \quad (7)$$

The GZZ combines the advantages of the Mogi and Hoek–Brown strength criteria. It has a high fitting accuracy for the true triaxial test data of rock. The GZZ can be simplified to the GHB at triaxial tensile and compressive states. Therefore, the parameter system of the Hoek–Brown criterion accumulated from experiment and engineering experience can be directly and conveniently used in the engineering application of the GZZ.

The derivation and analysis in this study were based primarily on the GZZ, and the calculation results were compared with those of the 2D GHB and circular GPH to determine the differences in the calculation results using different strength criteria.

2.2. Post-peak brittle softening behavior of rock mass

This study used elastic–brittle and elastic–perfectly plastic models for analysis. The post-peak mechanical behavior of the rock mass is

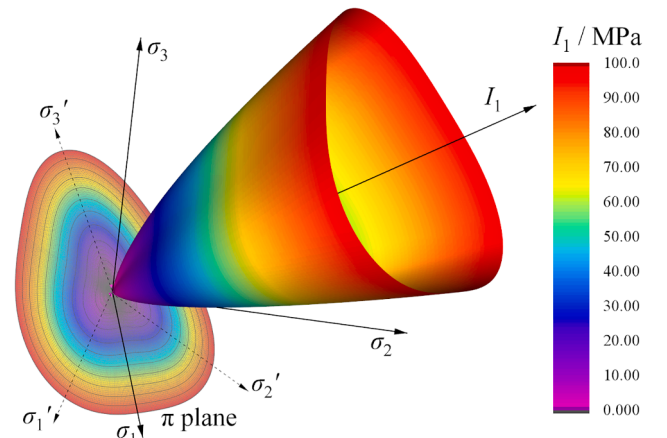


Fig. 3. Spatial envelope surface of the GZZ.

described by the GSI of the Hoek–Brown criterion parameter system. For the elastic–brittle model, the strength parameter immediately decreases to the residual value when the rock mass yields, as shown in Fig. 4. The residual value of the rock mass parameter is obtained using the following empirical formula (Cai et al., 2007).

$$GSI_r = GSI_0 \exp(-0.0134 GSI_0), \quad (8)$$

where GSI_0 is the initial value of the strength parameter. When the residual value is equal to the initial value, it is an elastic–perfectly plastic model.

In this process, the sudden change in the rock mass parameters results in a sudden change in the calculated stress. It is assumed that the radial stress remains continuous at the elastic–plastic interface (Sharan, 2003, 2005; Tu et al., 2018), whereas the circumferential and axial stresses can change abruptly, and the strain and displacement remain continuous.

3. Proposed solutions for the surrounding rock

3.1. Solution of the elastic zone

According to Lamé's solution of elastic mechanics, the stress in the elastic region can be expressed as:

$$\begin{cases} \sigma_r = p - (p - p_s)(R_0/r)^2, \\ \sigma_\theta = p + (p - p_s)(R_0/r)^2, \\ \sigma_z = q. \end{cases} \quad (9)$$

After removing the displacement caused by the initial ground stress, the strain and radial displacement of the surrounding rock can be expressed as:

$$\begin{cases} \varepsilon_r = (-p + p_s)R_0^2/(2Gr^2), \\ \varepsilon_\theta = (p - p_s)R_0^2/(2Gr^2), \\ u_r = (p - p_s)R_0^2/(2Gr), \end{cases} \quad (10)$$

where ε_θ and ε_r are the circumferential and radial strains of the tunnel, respectively; u_r is the radial displacement; $G = \frac{E}{2(1+\nu)}$ is the shear modulus of the surrounding rock; and ν is Poisson's ratio.

When the supporting pressure p_s is sufficient, the surrounding rock is in an elastic state, which can be solved using Eqs. (9) and (10). When $p_s < p_{cr}$, the surrounding rock near the tunnel wall will be in a plastic state. Additionally, the solution of the surrounding rock (still in the elastic zone) needs to take the component at the interface of the elastic–plastic zone as the boundary condition. Setting $R_p = R_0$ and $p_{cr} = p_s$, Eqs. (9) and (10) can be written as:

$$\begin{cases} \sigma_r = p - (p - p_{cr})(R_p/r)^2, \\ \sigma_\theta = p + (p - p_{cr})(R_p/r)^2, \\ \sigma_z = q, \\ \varepsilon_r = (-p + p_{cr})R_p^2/(2Gr^2), \\ \varepsilon_\theta = (p - p_{cr})R_p^2/(2Gr^2), \\ u_r = (p - p_{cr})R_p^2/(2Gr). \end{cases} \quad (11)$$

Eq. (11) is the solution of elastic zone when a plastic zone exits. It contains two unknowns, namely the radial stress p_{cr} and the radius R_p at the elastic–plastic interface. The two unknowns are given in the following solutions.

3.2. Solution at the elastic–plastic interface

The stress component of the elastic–brittle rock mass undergoes a sudden change at the elastic–plastic interface. In this study, the elastic–plastic interface is defined as the zero point. The stress on the elastic side is $\sigma_{(0-)}$ and the stress on the plastic side is $\sigma_{(0+)}$, as shown in Fig. 5.

The stress $\sigma_{(0-)}$ on the elastic side should satisfy both the stress expression of the elastic zone and the consistency condition expression of the plastic zone. Here, $r = R_p$, and the following equation can be obtained from Eq. (11).

$$\begin{cases} \sigma_{r(0-)} = p_{cr}, \\ \sigma_{\theta(0-)} = 2p - p_{cr}, \\ \sigma_{z(0-)} = q. \end{cases} \quad (12)$$

The tunnel axial stress σ_z is generally the intermediate principal stress and $\sigma_\theta > \sigma_z > \sigma_r$; that is, $\sigma_{(0-)} = [\sigma_1, \sigma_2, \sigma_3] = [\sigma_{\theta(0-)}, \sigma_{z(0-)}, \sigma_{r(0-)}] = [2p - p_{cr}, q, p_{cr}]$. Substituting this into Eq. (1), the following equation is obtained.

$$F = F(\sigma_{(0-)}) = F(p_{cr}) = 0. \quad (13)$$

At this time, only one unknown p_{cr} is included in the consistency condition, and the radial stress p_{cr} at the elastic–plastic interface can be obtained by solving Eq. (13).

The radial stress component σ_r of the elastic–brittle model is continuous at the elastic–plastic interface and the radial stress on the plastic side is still p_{cr} . When the strength parameters of the rock mass change, the values of σ_θ and σ_r calculated from the consistency condition produce mutations. For the solution of $\sigma_{(0+)} = p_{cr}$ is known, so the yield criterion is the equations of $\sigma_{\theta(0+)}$ and $\sigma_{z(0+)}$. A supplementary equation is needed to determine the stress components at this point, which is usually the supplementary equation of the axial stress.

To obtain the supplementary equation of the axial stress, Vrakas et al. (Chen et al., 2021; Vrakas and Anagnostou, 2014; Zou et al., 2016)

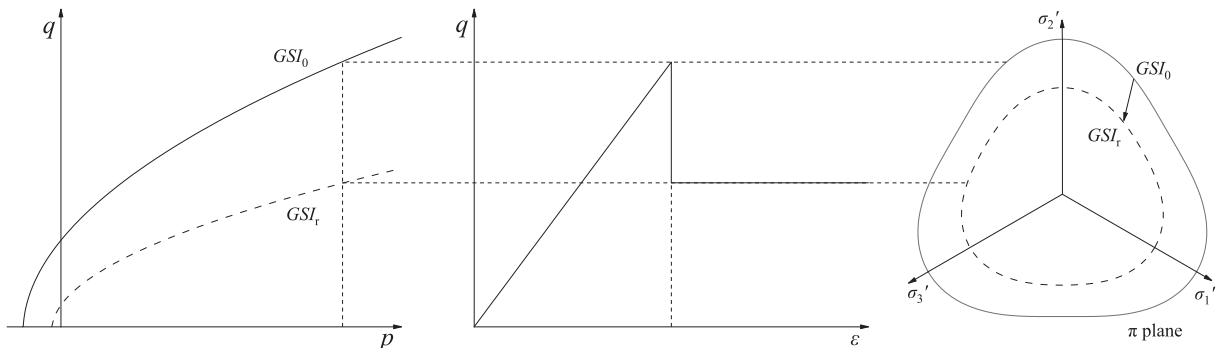


Fig. 4. Schematic of strength parameter drop.

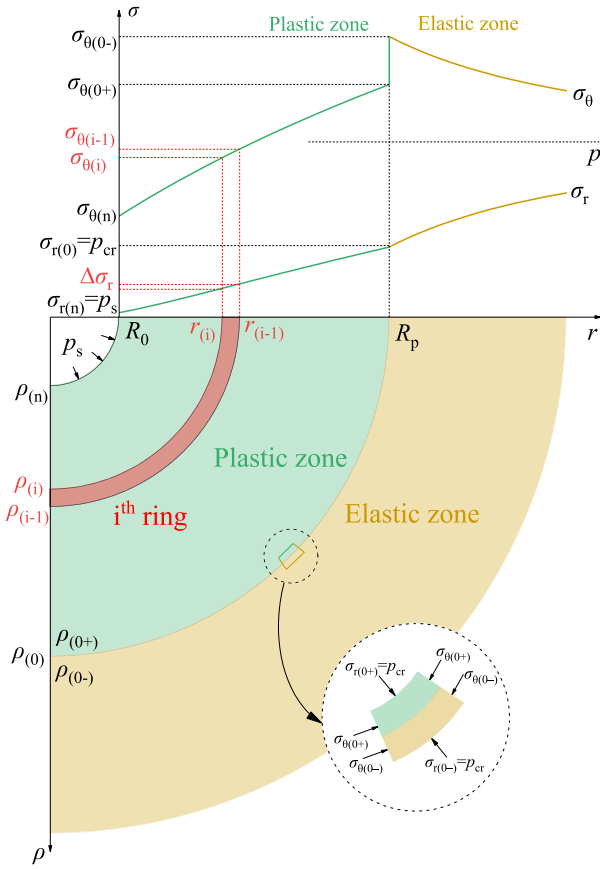


Fig. 5. Schematic of plastic zone division.

directly applied the intermediate principal stress formula derived from the Hooke's elastic law to the solution of the plastic zone (in the analysis process). The stress state and deformation process in the plastic zone are closely related to the rock dilatancy (Alejano and Alonso, 2005; Cai et al., 2022c; Walton et al., 2014; Zhao and Cai, 2010). They are not simply subject to Hooke's law. Deng et al. (2008) obtained the formula of the axial stress in the plastic zone using the uniform matching relationship between the Mohr–Coulomb strength and the Mises criteria:

$$\sigma_{z(i)} = \frac{1 - \sin\psi}{2} \sigma_{\theta(i)} + \frac{1 + \sin\psi}{2} \sigma_{r(i)}. \quad (14)$$

This method considers that the σ_z of the plastic zone is affected by the rock dilation angle ψ . For poor quality rock mass, it is considered that it obeys the law of constant volume flow (Hoek and Brown, 1997) and no volume expansion occurs during the excavation process, that is, $\psi = 0$, thus, Eq. (14) can be rewritten as $\sigma_{z(i)} = (\sigma_{\theta(i)} + \sigma_{r(i)})/2$.

Furthermore, $\sigma_{\theta(0+)}$ and $\sigma_{z(0+)}$ on the plastic side at the plastic–elastic interface can be obtained by combining the supplementary equations of the axial stress in Eq. (14) and the yield equation after changing the strength parameters.

The strain and displacement at the elastic–plastic interface are continuous. Hence, they can be obtained by introducing $\sigma_{\theta(0+)}$ into the strain and displacement solution in Eq. (11) in the elastic zone.

3.3. Stress of plastic zone

The stress of the plastic zone is difficult to solve directly and provide a strict analytical solution because the plastic zone consistency condition (1) is a transcendental equation. In this study, the approximate solution is given using the finite difference method with reference to Lee and Pietruszczak (2008).

The surrounding rock in the plastic zone is divided into n rings based

on an equal radial stress difference $\Delta\sigma_r = \sigma_{r(i)} - \sigma_{r(i-1)} = \frac{p_{cr} - p_s}{n}$, as shown in Fig. 5. The thickness of each ring $\Delta r_i = r_{(i-1)} - r_{(i)}$ is different. $i = 0$ and $i = n$ are the positions of the plastic–elastic interface and tunnel wall, respectively. The radii are normalized as:

$$\rho = r/R_p. \quad (15)$$

Similarly, $\rho_{(0)} = 1$ and $\rho_{(n)} = R_0/R_p$ are the relative radii of the plastic–elastic interface and tunnel wall, respectively.

The stress component calculation was conducted from the plastic–elastic interface to the tunnel wall. The initial condition when $i = 0$ is the solution of the stress and strain on the plastic side at the plastic–elastic interface, which can be obtained using Eq. (11):

$$\begin{Bmatrix} \sigma_{r(0)} \\ \sigma_{\theta(0)} \\ \sigma_{z(0)} \\ \varepsilon_{r(0)} \\ \varepsilon_{\theta(0)} \end{Bmatrix} = \begin{Bmatrix} p_{cr} \\ \sigma_{\theta(0+)} \\ \sigma_{z(0+)} \\ (p_{cr} - p)/(2G) \\ (p - p_{cr})/(2G) \end{Bmatrix}. \quad (16)$$

For the inside of the i -th ring ($\rho_{(i)}$), $\sigma_{r(i)} = \sigma_{r(0)} + i \cdot \Delta\sigma_r$ is known and only two unknowns, i.e., $\sigma_{\theta(i)}$ and $\sigma_{z(i)}$, must be solved. For the GHB, $\sigma_{z(i)}$ is independent of the yield criterion and can be separated from the formula to obtain $\sigma_{\theta(i)} = \sigma_{r(i)} + \sigma_c (m_i \sigma_{r(i)} / \sigma_c + s)^a$. However, the yield equation of the 3D Hoek–Brown criterion (GPH and GZZ) cannot separate $\sigma_{\theta(i)}$ to obtain its value directly. The common solution approach (Xia et al., 2018; Xu et al., 2021) is to add a supplementary equation of the axial stress to eliminate $\sigma_{z(i)}$ and directly solve the yield surface equation. This helps obtain the unknown quantity $\sigma_{\theta(i)}$, and subsequently obtain $\sigma_{z(i)}$. An iterative solution, such as the Newton–Raphson method or directly calling MATLAB's built-in functions, is generally required because the yield criterion is a transcendental equation. Solving n $\sigma_{\theta(i)}$ iteratively is time consuming. This study proposes a new and simple approach for solving $\sigma_{(i)}$.

The outside ($\rho_{(i-1)}$) and inside ($\rho_{(i)}$) of the i -th ring should satisfy the plastic zone consistency condition; that is, $F(\sigma_{(i)}) = F(\sigma_{(i-1)}) = 0$. If the outside stress of the i -th ring $\sigma_{(i-1)} = (\sigma_{r(i-1)}, \sigma_{\theta(i-1)}, \sigma_{z(i-1)})$ is a known quantity, that is, $F(\sigma_{(i-1)}) = 0$ is established, then the equivalent equation for solving the inside stress of the i -th ring $F(\sigma_{(i)}) = 0$ is.

$$dF = F(\sigma_{(i)}) - F(\sigma_{(i-1)}) = 0. \quad (17)$$

For the straight section of the elastic–brittle and elastic–perfect plastic models, the following equation can be obtained from the consistency conditions on the yield surface (Lubliner, 2008).

$$dF = \frac{\partial F}{\partial \sigma_{ij}} d\sigma_{ij} = \frac{\partial F}{\partial \sigma_r} d\sigma_r + \frac{\partial F}{\partial \sigma_{\theta}} d\sigma_{\theta} + \frac{\partial F}{\partial \sigma_z} d\sigma_z = 0. \quad (18)$$

Substituting $d\sigma_r \approx \sigma_{r(i)} - \sigma_{r(i-1)}$, $d\sigma_{\theta} \approx \sigma_{\theta(i)} - \sigma_{\theta(i-1)}$ and $d\sigma_z \approx \sigma_{z(i)} - \sigma_{z(i-1)}$ using the difference method, and $\frac{\partial F}{\partial \sigma_k} = \frac{\partial F}{\partial \sigma_{k(i-1)}} (k = r, \theta, z)$ into Eq. (18) to obtain the linear expressions of $\sigma_{(i)}$ and $\sigma_{(i-1)}$:

$$\frac{\partial F}{\partial \sigma_{r(i-1)}} (\sigma_{r(i)} - \sigma_{r(i-1)}) + \frac{\partial F}{\partial \sigma_{\theta(i-1)}} (\sigma_{\theta(i)} - \sigma_{\theta(i-1)}) + \frac{\partial F}{\partial \sigma_{z(i-1)}} (\sigma_{z(i)} - \sigma_{z(i-1)}) = 0. \quad (19)$$

The coefficient $\frac{\partial F}{\partial \sigma_{k(i-1)}} (k = r, \theta, z)$ can be directly obtained because $\sigma_{(i-1)} = (\sigma_{r(i-1)}, \sigma_{\theta(i-1)}, \sigma_{z(i-1)})$ is known. The calculation coefficient based on the GZZ can be obtained from Eq. (B.6) in Appendix B.

Combining the two linear Eqs. (19) and (14), they can be written as linear expressions:

$$\begin{bmatrix} c_{\theta 1} & c_{z 1} \\ c_{\theta 2} & c_{z 2} \end{bmatrix} \begin{bmatrix} \sigma_{\theta(i)} \\ \sigma_{z(i)} \end{bmatrix} = \begin{bmatrix} L_1 \\ L_2 \end{bmatrix}, \quad (20)$$

where,

$$\begin{cases} c_{\theta 1} = \frac{\partial F}{\partial \sigma_{\theta(i-1)}}, c_{z1} = \frac{\partial F}{\partial \sigma_{z(i-1)}}, \\ c_{\theta 2} = \frac{1 - \sin \psi}{2}, c_{z2} = -1, \\ L_1 = \frac{\partial F}{\partial \sigma_{\theta(i-1)}} \sigma_{\theta(i-1)} + \frac{\partial F}{\partial \sigma_{z(i-1)}} \sigma_{z(i-1)} - \frac{\partial F}{\partial \sigma_{r(i-1)}} \Delta \sigma_r, \\ L_2 = -\frac{1 + \sin \psi}{2} \sigma_{r(i)}. \end{cases} \quad (21)$$

For binary linear equations, their explicit expressions can be directly written as.

$$\begin{cases} \sigma_{\theta(i)} = \frac{L_1 c_{z2} - L_2 c_{z1}}{c_{\theta 1} c_{z2} - c_{\theta 2} c_{z1}}, \\ \sigma_{z(i)} = \frac{L_1 c_{\theta 2} - L_2 c_{\theta 1}}{c_{z1} c_{\theta 2} - c_{z2} c_{\theta 1}}. \end{cases} \quad (22)$$

Eq. (22) directly provides the explicit expressions of $\sigma_{\theta(i)}$ and $\sigma_{z(i)}$. Unlike conventional methods, the process of iteratively solving

$$\begin{bmatrix} \Delta \epsilon_{r(i)}^e \\ \Delta \epsilon_{\theta(i)}^e \end{bmatrix} = \frac{1}{2G} \begin{bmatrix} 1 - \nu & -\nu \\ -\nu & 1 - \nu \end{bmatrix} \begin{bmatrix} \Delta \sigma_{r(i)} \\ \Delta \sigma_{\theta(i)} \end{bmatrix} = \frac{1}{2G} \begin{bmatrix} 1 - \nu & -\nu \\ -\nu & 1 - \nu \end{bmatrix} \begin{bmatrix} \sigma_{r(i)} - \sigma_{r(i-1)} \\ \sigma_{\theta(i)} - \sigma_{\theta(i-1)} \end{bmatrix}. \quad (29)$$

nonlinear equations can be avoided and the calculation rate can be significantly improved. This method can be applied to the GZZ and other strength criteria. The calculation coefficients $\frac{\partial F}{\partial \sigma_{k(i-1)}} (k = r, \theta, z)$ for different strength criteria are different. The derivation process and final expression of the calculation coefficients of the three strength criteria of the GZZ, GPH, and GHB are given in Appendix B.

All the stress components of $i = 1-n$ can be calculated step by step according to the initial conditions $\sigma_{(0)}$ at the elastic-plastic interface given in Eq. (16).

3.4. Radius of each point in the plastic zone

The stress component in the plastic zone should satisfy the equilibrium equation (Lubliner, 2008):

$$\frac{d\sigma_r}{dr} - \frac{\sigma_{\theta} - \sigma_r}{r} + F_r = 0, \quad (23)$$

where F_r is the volume force in the radial direction of the surrounding rock, which is the stress component of gravity in the radial direction. $F_r = 0$ because the tunnel scale is smaller than the stratum scale. For normalizing the radius, the difference format of Eq. (23) is given as:

$$\frac{\Delta \sigma_r}{\rho_{(i)} - \rho_{(i-1)}} - \frac{\sigma_{\theta(i)} - \sigma_{r(i)}}{\rho_{(i)}} = 0. \quad (24)$$

By rearranging Eq. (24) and combining the obtained stress components $\sigma_{(i)}$ and $\sigma_{(i-1)}$, the radius inside the i -th ring can be obtained as:

$$\rho_{(i)} = \rho_{(i-1)} \frac{\sigma_{\theta(i)} - \sigma_{r(i)}}{\sigma_{\theta(i)} - \sigma_{r(i)} - \Delta \sigma_r} = \rho_{(0)} \prod_{j=1}^i \frac{\sigma_{\theta(j)} - \sigma_{r(j)}}{\sigma_{\theta(j)} - \sigma_{r(j)} - \Delta \sigma_r}. \quad (25)$$

The initial condition of Eq. (25) is $\rho_{(0)} = 1$ at the elastic-plastic interface, and all the radii of $i = 1-n$ can be gradually calculated.

3.5. Strain and deformation of the plastic zone

For the i -th ring, its inside strain component $\epsilon_{(i)}$ can be divided into

elastic and plastic strains as:

$$\begin{Bmatrix} \epsilon_{r(i)} \\ \epsilon_{\theta(i)} \end{Bmatrix} = \begin{Bmatrix} \epsilon_{r(i)}^e \\ \epsilon_{\theta(i)}^e \end{Bmatrix} + \begin{Bmatrix} \epsilon_{r(i)}^p \\ \epsilon_{\theta(i)}^p \end{Bmatrix}. \quad (26)$$

The elastic strain can be obtained from the elastic strain outside the i -th ring and the corresponding elastic strain increment as:

$$\begin{Bmatrix} \epsilon_{r(i)}^e \\ \epsilon_{\theta(i)}^e \end{Bmatrix} = \begin{Bmatrix} \epsilon_{r(i-1)}^e \\ \epsilon_{\theta(i-1)}^e \end{Bmatrix} + \begin{Bmatrix} \Delta \epsilon_{r(i)}^e \\ \Delta \epsilon_{\theta(i)}^e \end{Bmatrix}. \quad (27)$$

When $i = 0$, the calculation point is at the elastic-plastic interface and the plastic strain is $\epsilon_{(0)}^p = 0$; therefore, the initial condition of Eq. (27) can be obtained using Eq. (16):

$$\begin{Bmatrix} \epsilon_{r(0)}^e \\ \epsilon_{\theta(0)}^e \end{Bmatrix} = \begin{Bmatrix} \epsilon_{r(0)} \\ \epsilon_{\theta(0)} \end{Bmatrix} - \begin{Bmatrix} \epsilon_{r(0)}^p \\ \epsilon_{\theta(0)}^p \end{Bmatrix} = \begin{Bmatrix} \epsilon_{r(0)} \\ \epsilon_{\theta(0)} \end{Bmatrix} = \begin{Bmatrix} (p_{cr} - p)/(2G) \\ (p - p_{cr})/(2G) \end{Bmatrix}. \quad (28)$$

The elastic strain increment can be obtained using Hooke's law:

The plastic strain component can be obtained as follows.

The plastic zone strain should satisfy the compatibility equation (Lubliner, 2008).

$$\frac{d\epsilon_{\theta}}{dr} + \frac{\epsilon_{\theta} - \epsilon_r}{r} = 0. \quad (30)$$

The following equation can be obtained after elastic-plastic strain separation.

$$\frac{d\epsilon_{\theta}^p}{dr} + \frac{\epsilon_{\theta}^p - \epsilon_r^p}{r} = -\frac{d\epsilon_{\theta}^e}{dr} - \frac{\epsilon_{\theta}^e - \epsilon_r^e}{r}. \quad (31)$$

The following equation can be obtained from Eq. (31) for the inside and outside of the i -th ring.

$$\frac{\epsilon_{\theta(i)}^p - \epsilon_{\theta(i-1)}^p}{r_{(i)} - r_{(i-1)}} + \frac{\epsilon_{\theta(i)}^p - \epsilon_{r(i)}^p}{r_{(i)}} = -\frac{\epsilon_{\theta(i)}^e - \epsilon_{\theta(i-1)}^e}{r_{(i)} - r_{(i-1)}} - \frac{\epsilon_{\theta(i)}^e - \epsilon_{r(i)}^e}{r_{(i)}}. \quad (32)$$

Considering the non-associated flow rule related to the dilatancy characteristics of the rock, it is assumed that the radial plastic strain and the circumferential plastic strain satisfy.

$$d\epsilon_r^p + \lambda d\epsilon_{\theta}^p = 0, \quad (33)$$

where λ is the dilatation coefficient, and its relationship with the dilation angle ψ is.

$$\lambda = \frac{1 + \sin \psi}{1 - \sin \psi}. \quad (34)$$

When the dilatation angle $\psi = 0$, a dilatation coefficient of $\lambda = 1$ indicates that the plastic volume strain is not considered.

The difference format of Eq. (33) is.

$$\epsilon_{r(i)}^p - \epsilon_{r(i-1)}^p + \lambda (\epsilon_{\theta(i)}^p - \epsilon_{\theta(i-1)}^p) = 0. \quad (35)$$

The plastic strain obtained by combining Eqs. (32) and (35) is expressed as:

$$\begin{cases} \varepsilon_{\theta(i)}^p = \frac{-1}{1 + (1 + \lambda)\eta_{(i)}} \left[\left(\varepsilon_{\theta(i)}^e - \varepsilon_{\theta(i-1)}^e \right) + \eta_{(i)} \left(\varepsilon_{\theta(i)}^e - \varepsilon_{r(i)}^e \right) - \varepsilon_{\theta(i-1)}^p - \eta_{(i)} \left(\lambda \varepsilon_{\theta(i-1)}^p + \varepsilon_{r(i-1)}^p \right) \right], \\ \varepsilon_{r(i)}^p = \varepsilon_{r(i-1)}^p - \lambda \left(\varepsilon_{\theta(i)}^p - \varepsilon_{\theta(i-1)}^p \right), \end{cases} \quad (36)$$

where,

$$\begin{cases} \eta_{(i)} = 1 - r_{(i-1)}/r_{(i)}, \\ \left(\varepsilon_{\theta(i)}^e - \varepsilon_{\theta(i-1)}^e \right) = \frac{1}{2G} \left[(1 - \nu)(\sigma_{\theta(i)} - \sigma_{\theta(i-1)}) - \nu(\sigma_{r(i)} - \sigma_{r(i-1)}) \right], \\ \left(\varepsilon_{\theta(i)}^e - \varepsilon_{r(i)}^e \right) = \frac{\sigma_{\theta(i)} - \sigma_{r(i)}}{2G}. \end{cases} \quad (37)$$

The initial condition, $i = 0$ in Eq. (36) is at the elastic–plastic interface, and the plastic strain at all positions $i = 1-n$ can be calculated step by step.

The strain components at all positions can be obtained by substituting Eq. (27) for the elastic strain and Eq. (36) for the plastic strain into Eq. (26).

After n calculations, the stress–strain components and the relative radii of all positions in the plastic zone are obtained. The radius R_p of the elastic–plastic interface can be obtained by substituting $\rho_{(n)}$ into Eq. (15), as follows.

$$R_p = R_0/\rho_{(n)}. \quad (38)$$

The unknowns of the stress–strain and displacement solutions in the elastic region have been solved completely, and the solutions of the relevant components in the elastic region can be obtained.

The radius of each point i in the plastic zone can be calculated based on the relative radius R_p :

$$r = \rho R_p. \quad (39)$$

For an axisymmetric state around the opening, the geometric equations in the cylindrical coordinate system can be written in Euler form, as follows.

$$\varepsilon_{\theta} = \ln \left(1 + \frac{u_r}{r} \right) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} \left(\frac{u_r}{r} \right)^{n+1} \approx \frac{u_r}{r}. \quad (40)$$

Thus, the displacement of each point can be obtained using the

following equation:

$$u_r = \varepsilon_{\theta} r. \quad (41)$$

The aforementioned calculation process is presented in Appendix A.

4. Verification of the proposed solution

To verify the method proposed in this study, the calculation results were compared with those of the conventional method, nonlinear equations using MATLAB's built-in *fsolve* function, and 3D finite element analysis software GeoFBA^{3D}. The GZZ strength criterion and the elastic–perfect plastic model were used in the calculation. The rock mass parameters were: $GSI_0 = GSI_r = 50$, $m_i = 6$, $D = 0$, $\sigma_c = 40$ MPa, $E = 2$ GPa, $\nu = 0.35$, $\psi = 10^\circ$, excavation radius $R_0 = 8$ m, $p = q = 30$ MPa, $p_s = 1$ MPa, and $n = 1000$.

The finite element calculation model takes a quarter of circle and only the circumferential displacement constraint is applied on the symmetry-plane. A displacement constraint is applied on one side perpendicular to the paper-plane, and the axial stress q is applied on the other side. The initial stress p is applied to the inner and the outer radius. After the initial stress is balanced, the displacement is cleared and the pressure applied to the inner radius is replaced to the support stress p_s . The outer boundary radius of the model is 120 m, as shown in Fig. 6.

Fig. 7 shows the stress and displacement of the surrounding rock calculated using different methods. It can be observed from the figure that the calculation results of the proposed method are almost consistent with those of the conventional iterative solution method. The results are correlated with those of the 3D finite element numerical calculation, which validates the accuracy of the proposed method.

A computer with an Intel(R) i7-7700 CPU @ 3.60 GHz and 32.0 GB memory was used to test the calculation efficiency. The calculation time for solving the stress and strain in the plastic zone was recorded. It can be seen from Fig. 8 that the calculation time of the two methods increased linearly with an increase in the division number n , whereas the calculation time of the proposed method was significantly less than that of the conventional method. When $n \geq 500$, the computational efficiency of the proposed method was >12,000 times that of the conventional method.

When $n = 1000$, all the calculation times, including the calculation of the elastic zone, plastic zone, and elastic–plastic interface, were recorded. The calculation time of the conventional method was 6.977 s (6.400 s for the plastic zone, accounting for 91.7 %) and that of the proposed method was 0.105 s (0.005 s for the plastic zone, accounting for only 4.7 %). Although other common calculation times limit the advantages of the proposed method in terms of the overall calculation efficiency, the overall calculation efficiency ratio was still as high as 66.45.

In terms of the calculation accuracy, the tunnel wall displacement u_{R0} and plastic zone radius R_p at different n conditions were calculated. The results obtained using the proposed method at $n = 10^8$ were taken as the true value. Additionally, the calculation errors of the two methods were compared. It can be seen from Fig. 9 that as n increased, the u_{R0} obtained using the two methods gradually decreased and converged to an exact value, whereas the results for R_p were different. The R_p obtained using the conventional method gradually decreased with an increase in n , whereas that of the proposed method gradually increased; however, both converged to the same value. The sensitivity of u_{R0} to n

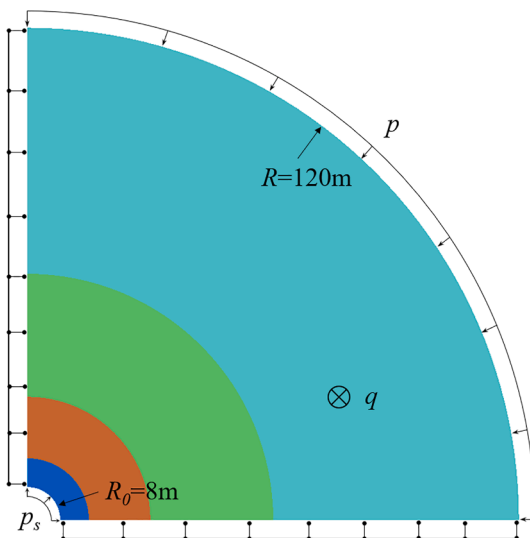


Fig. 6. Finite element calculation model.

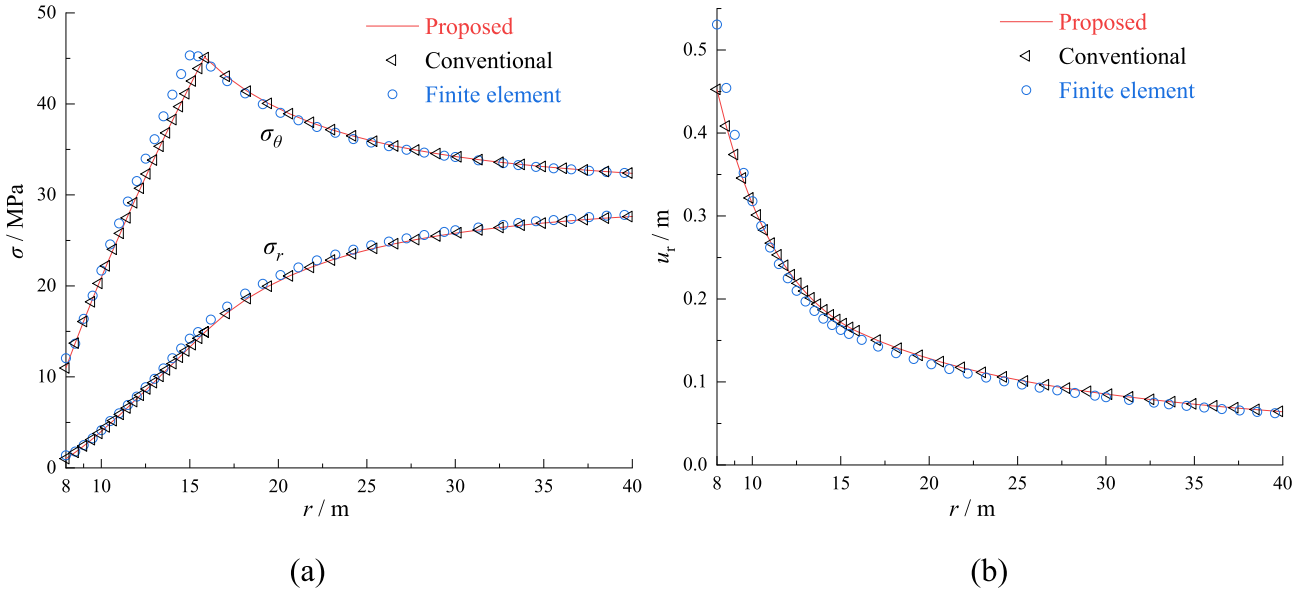


Fig. 7. Results of the surrounding rock obtained using different methods: (a) stress; and (b) displacement.

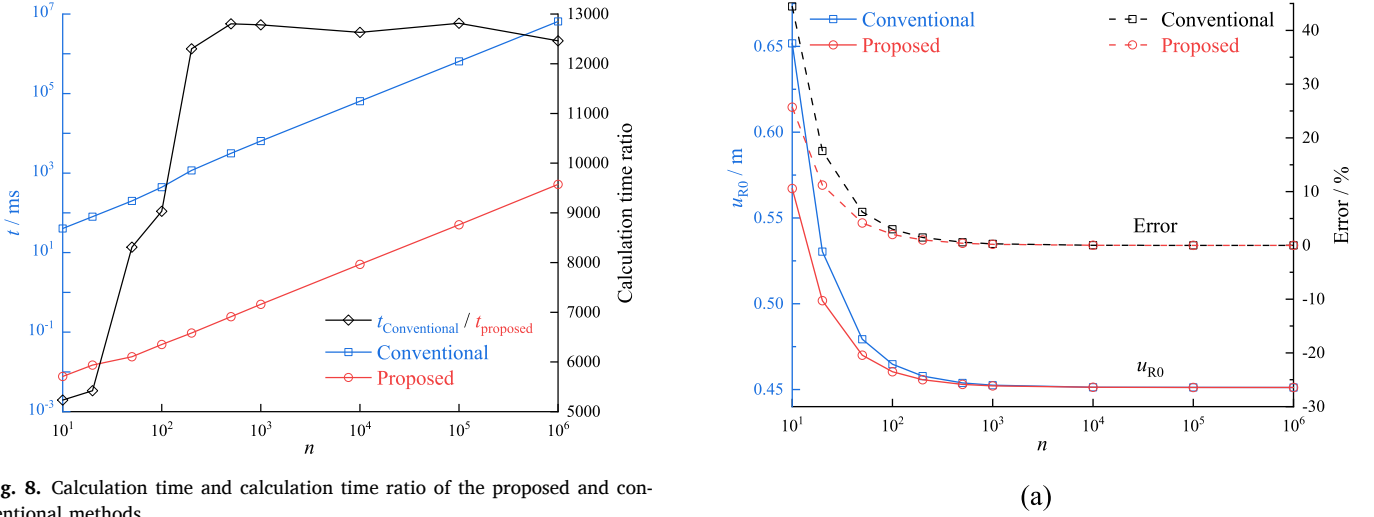


Fig. 8. Calculation time and calculation time ratio of the proposed and conventional methods.

was higher. At $n = 10$, the calculation error of the conventional method was up to 44 %, whereas that of the proposed method was up to 26 %, and the error gradually decreased with an increase in n . The sensitivity of R_p to n was lower. At $n = 10$, the calculation errors of the two methods were only 2.3 and 1.2 %, respectively. The calculation error of the proposed method was lower than that of the conventional method for all values of n . The error ratio of the two methods, u_{R0} was less than 0.69, and R_p was less than 0.60. When $n \geq 200$, both the errors of u_{R0} and R_p of proposed method were less than 1.0 %, which is an acceptable accuracy and can meet most calculation requirement. The proposed method also had significant advantages in terms of its calculation accuracy.

The above difference between the two methods came from the processing of the difference instead of the differential. The stress result solved by the conventional method was accurate. When dealing with the equilibrium equation, using the difference instead of the differential made the first term of Eq. (24) larger and the obtained R_p larger, thereby resulting in a significant error, as shown in Fig. 9(b). When dealing with Eq. (18) in the proposed method, the use of difference instead of differential made the obtained stress σ_θ larger, that is, it rendered the second term of Eq. (24) larger. The errors of the first and the second terms of Eq. (24) could be partially offset, so that the final R_p was

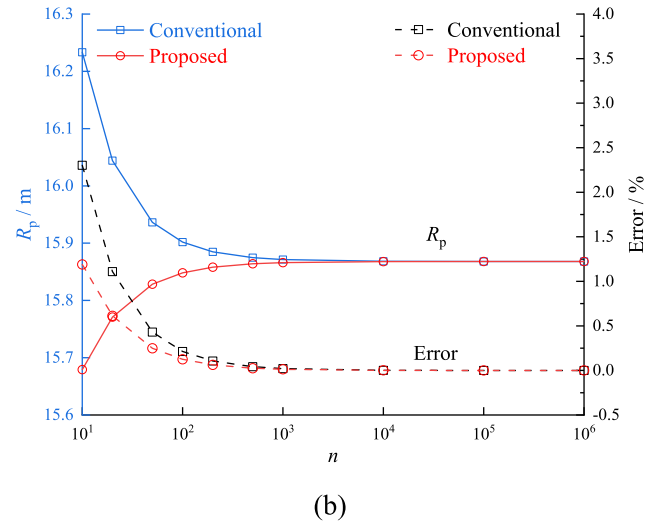


Fig. 9. Results comparison of the proposed and conventional methods: (a) displacement of the tunnel wall; and (b) radius of the plastic zone.

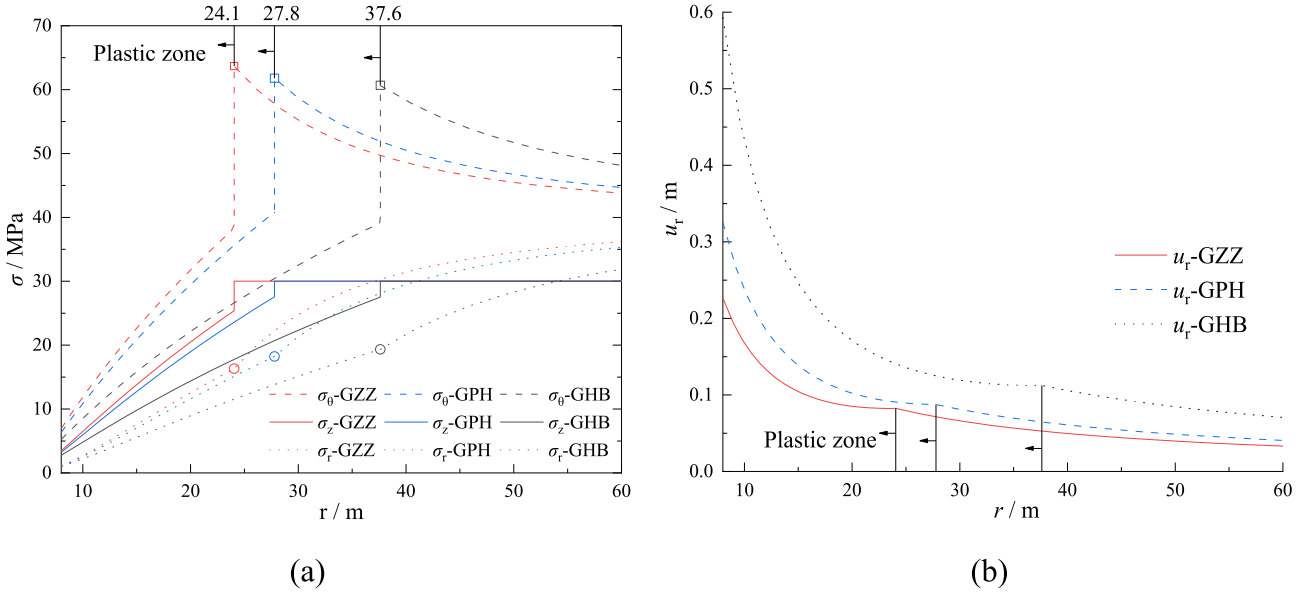


Fig. 10. Comparison of results calculated using different strength criteria: (a) stress; and (b) displacement.

smaller, as shown in Fig. 9(b). The error of the proposed method was smaller. Furthermore, the error in the displacement was also smaller as compared to that of the conventional method.

When the accuracy requirements were met (the maximum error was less than 0.5 %, $n = 1000$), the calculation time of the conventional method was only a few seconds (6.8 s), which appears to be very low. However, this was only the calculation time required to obtain the results of one set of parameters. In practical applications, the calculation results of multiple sets of parameters are often required, and the conventional method requires a long calculation time. For instance, it was necessary to calculate the u_{R0} obtained using different p and q values as shown in Fig. 12. The proposed method took 19.8 min, whereas the conventional method took 21.9 h. The overall computational efficiency ratio was 66.36.

5. Case analysis and discussion

5.1. Comparison of different strength criteria

To evaluate the differences in the calculation results of different yield criteria, we performed the calculation using the GZZ, GPH, and GHB, and the elastic–brittle model. The parameters of harder rock masses for

the elastic–brittle model were taken as: $GSI_0 = 65$, $GSI_r = 27.2$, calculated using Eq. (8), $m_i = 6$, $D = 0$, $\sigma_c = 50$ MPa, $E = 9$ GPa, $\nu = 0.3$, $\psi = 10^\circ$, excavation radius $R_0 = 8$ m, $p = 40$ MPa, $q = 30$ MPa, $p_s = 1$ MPa, and $n = 1000$.

Fig. 10 shows the results of the surrounding rock stress and displacement calculated using different strength criteria. At the elastic–plastic interface, the stress calculation before the drop was based on both the elastic equation ($I_1 = 110$ MPa) and the plastic yield criterion; the obtained result was $\sigma_{\theta GZZ} > \sigma_{\theta GPH} > \sigma_{\theta GHB}$, and the result of the GZZ was the largest, as shown in Fig. 10(a). In the plastic zone, the stress values calculated using the GZZ were always the highest and those of the GHB were the lowest, which was consistent with the magnitude of σ_θ before the stress drop. The plastic zone radii calculated using the GZZ, GPH, and GHB were 24.1, 27.8, and 37.6 m, respectively. The calculation result of the GZZ was the smallest. The slope of the displacement curve also changed at the interface because of the sudden change in stress at the elastic–plastic interface, as shown in Fig. 10(b), whereas the displacement value remained continuous. The corresponding calculated results of the tunnel wall displacement u_{R0} were 0.227, 0.325, and 0.593 m, respectively. The magnitude of the displacement results was consistent with the radii of the plastic zone, and the calculated result of the GZZ was the smallest.

From the aforementioned analysis, it was observed that the stability of the surrounding rock calculated using the GZZ was the best with the smallest plastic zone radius, the largest stress value, and the smallest displacement value, followed by that of the GPH, and the GHB was the worst. This was because the envelopes of the three strength criteria in the π -plane area (where the elastic stress path of this set of cases passed) had the same magnitude relationship, as shown by curve A in Fig. 11. The stress calculation in the elastic region was independent on the plastic yield criterion, and the elastic stress path was the same before the yields. With an increase in the elastic stress, the stress path moved to the right and the rock mass yielded when the yield surface was reached. Different yield criteria had different yield surface locations and stress levels at the yield point. At the elastic stress path of this set of cases, the yield surface of the GZZ was the largest and rock mass strength was the highest; therefore, the stress level of the yield point was the highest and the stability of the surrounding rock was the best.

We calculated two sets of cases for comparison because the magnitude relationship of different yield criteria at different positions in the π -plane was different, as shown by curves B and C in Fig. 11. The $I_1 = 110$ MPa value of the three sets of cases was the same, hence, the yield

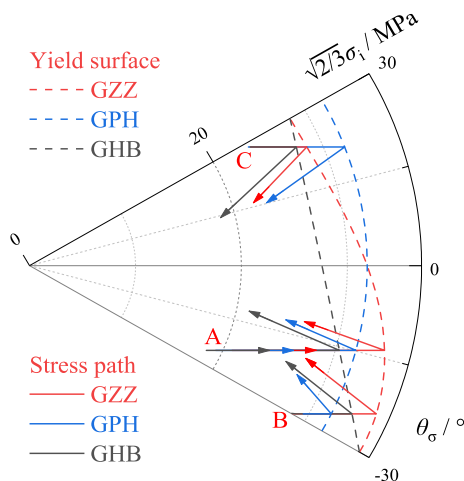


Fig. 11. Stress drop path at the elastic–plastic interface.

surface was the same, as shown by the dotted line in Fig. 11.

The difference in the calculation results of the different yield criteria was closely related to the relative position of the yield surface in the π -plane. The stress level of the yield point of GZZ was always higher than that of GHB. This was because the yield surface of the GZZ was always at the periphery of the GHB. However, the GPH with a circular envelope had an intersecting relationship with the other two criteria, which was lower than that of the GHB near $\theta_\sigma = -\pi/6$ (curve B), higher than the GZZ at $\theta_\sigma > 0$ (curve C), and between them around $\theta_\sigma < 0$ (curve A). The stress level at the yield point of the GPH had an indeterminate magnitude as compared with those of the other two yield criteria. In addition, the yield stress level $\sqrt{2/3}\sigma_1 = 31.86$ MPa of the GPH criterion at different stress conditions was a fixed value, whereas the yield strength of the actual rock mass at different tensile and compressive conditions was different.

Unlike the 2D GHB, the GZZ (the 3D Hoek–Brown criterion) considered the effect of the intermediate principal stress σ_2 . The calculated radius and displacement of the plastic zone were smaller. The results showed that the intermediate principal stress σ_2 had an inhibitory effect on the development of the plastic zone and deformation of the surrounding rock, which has been theoretically and experimentally verified (Cai, 2008; Pan et al., 2012; Xu and Yu, 2006). However, the GPH (the 3D Hoek–Brown criterion) had an intersecting relationship with the 2D GHB. The magnitude relationship of the calculation results of the two was uncertain. The GPH underestimated the triaxial compressive strength of the rock mass. The yield point stress level obtained using the GPH was lower than that of the 2D GHB near $\theta_\sigma = -\pi/6$, as shown by curve B in Fig. 11, which indicates that the calculated stability of the surrounding rock considering σ_2 was worse and opposite results were obtained. In addition, the GPH overestimated the triaxial tensile strength of the rock mass when $\theta_\sigma > 0$. The yield point stress level obtained using the GPH was higher than that of GHB and even higher than that of GZZ, as shown by curve C in Fig. 11. Overestimating the strength of the rock mass led to an over-optimistic support design, and insufficient support may cause engineering disasters.

In practice, if the 2D GHB is used in the design, the effect of the intermediate principal stress cannot be considered and the calculated results will be conservative. If the circular GPH is used in the design, the rock mass strength cannot be accurately reflected and the obtained results will be conservative or too optimistic, with significant deviations, thereby resulting in an incorrect engineering design. Accurate analysis results require appropriate 3D strength theory, especially for deep tunnels with a wide distribution of plastic zones (Cai et al., 2022a; Cai et al., 2022b).

5.2. Parameter analysis

5.2.1. Effect of the axial stress q

Several studies have demonstrated that the intermediate principal stress has a significant effect on the stability of the surrounding rock (Cai, 2008; Feng et al., 2019; Pan et al., 2012; Song et al., 2019; Zhao et al., 2021); however, the effect of the initial axial stress on the stability of the surrounding rock has not yet been elucidated. To study the effect of the axial stress q on the calculation results, we used the elastic–brittle model and parameters presented in Section 5.1, and the GZZ and GPH in the calculation. The 2D GHB was independent of the axial stress q ; therefore, variations in q did not affect the results of the GHB.

It can be observed from Fig. 12(a) that the tunnel wall displacement u_{R0} and plastic zone radius R_p obtained using the GZZ and GPH first decreased and then increased with an increase in the axial stress q . There was an optimal initial axial stress value q_{opt} that yielded the smallest calculated results of u_{R0} and R_p and the best stability of the surrounding rock. The values of q_{opt} obtained using different strength criteria were different. For $q_{opt} = 35.8$ m obtained using the GZZ was slightly lower than the initial plane stress p , whereas the $q_{opt} = 49.6$ m obtained using

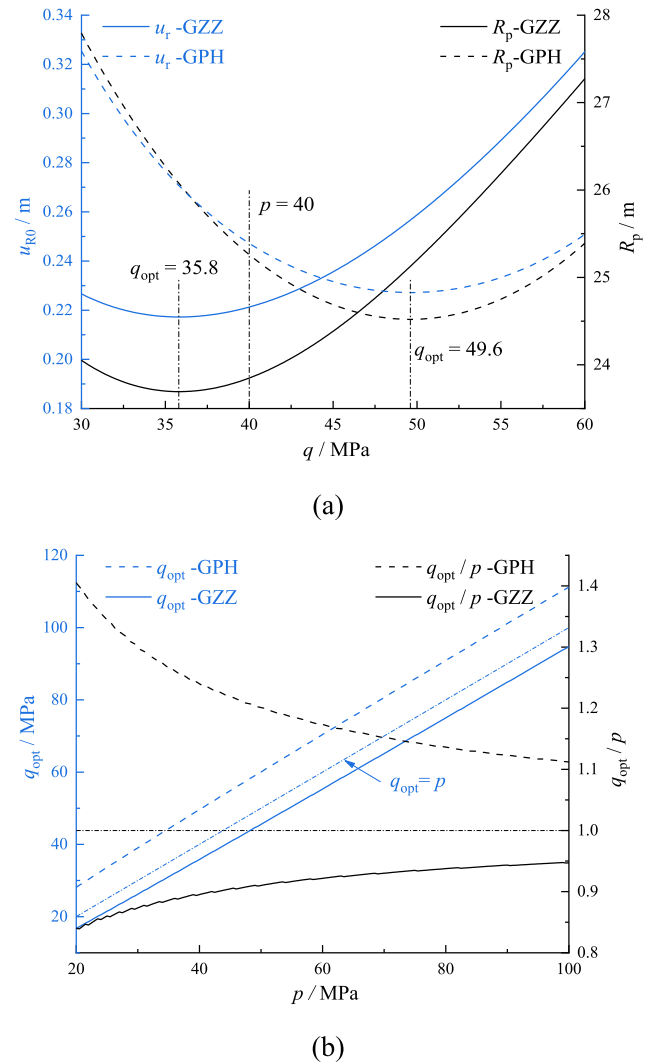


Fig. 12. (a) u_{R0} and R_p at different q conditions; and (b) q_{opt} at different p conditions.

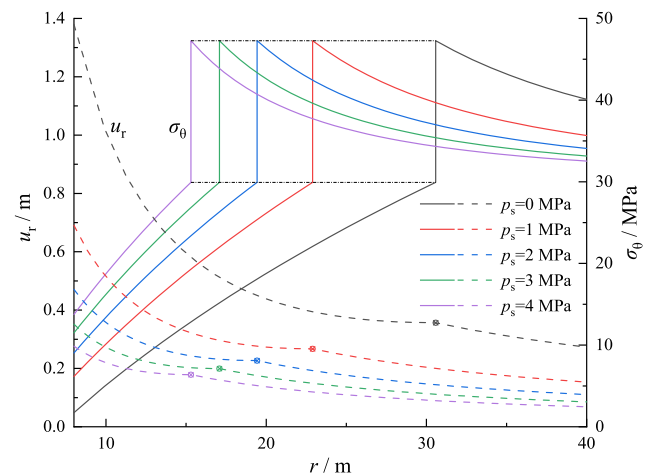


Fig. 13. Circumferential stress and radial displacement at different p_s conditions.

the GPH was slightly higher than p . The q_{opt} obtained using the two different strength criteria gradually increased with the increase in p . The ratio of q_{opt} to p gradually approached 1.0; that is, the optimal initial

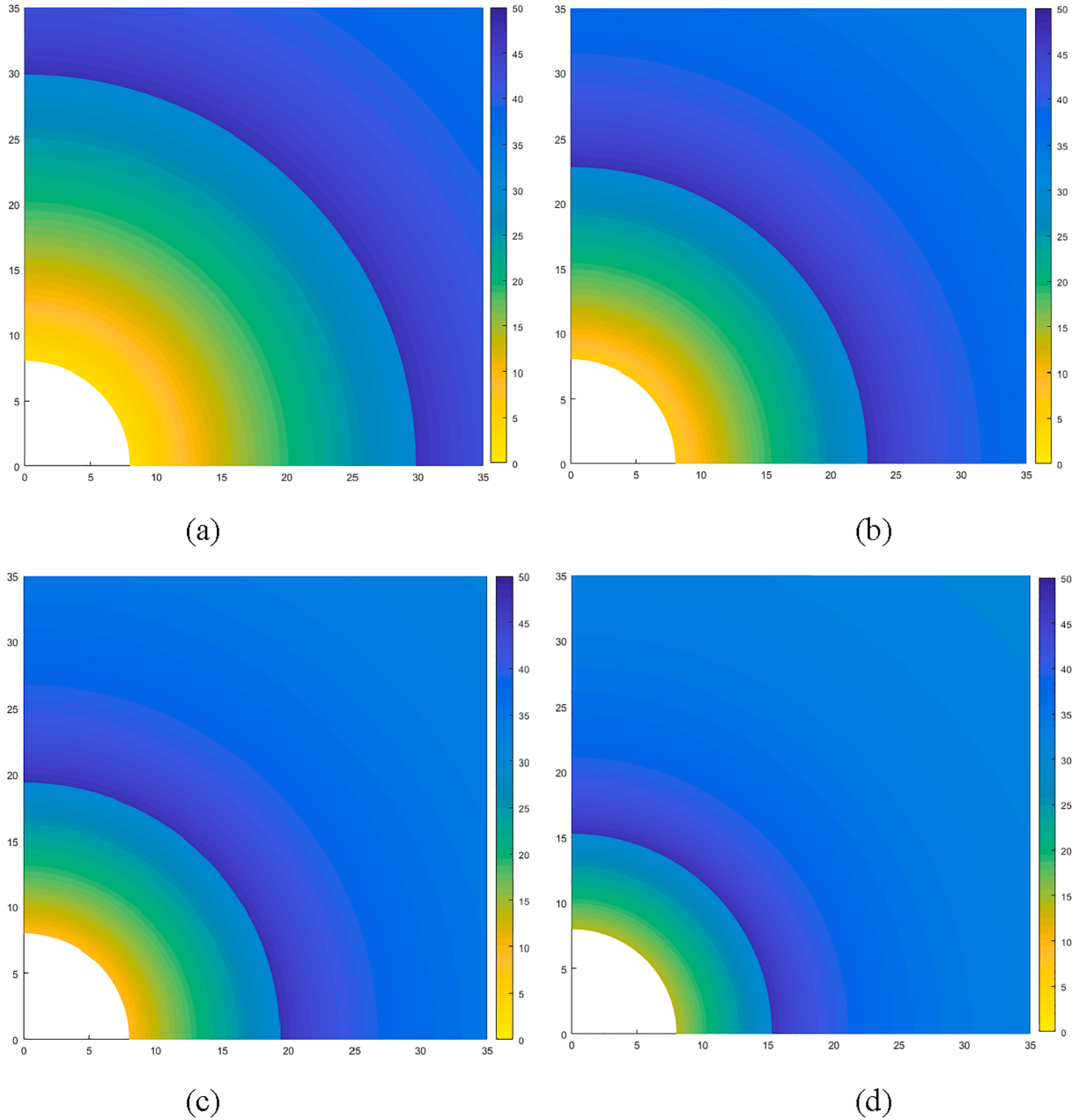


Fig. 14. Distribution of σ_θ at different p_s conditions: (a) $p_s = 0$ MPa; (b) $p_s = 1$ MPa; (c) $p_s = 2$ MPa; and (d) $p_s = 4$ MPa.

axial stress q_{opt} gradually approached the initial stress p , as shown in Fig. 12(b). The difference between the two results was owing to the difference in the p_{cr} values obtained using different yield equations, and the change in p_{cr} with q had the same change law as u_{R0} and R_p .

As presented in Section 5.1, the GPH could not accurately reflect the different tensile and compressive characteristics of geotechnical materials, and the validity of the obtained results was questionable. The 2D GHB ignored the effect of the intermediate principal stress and did not consider the effect of the axial stress on the stability of the surrounding rock; therefore, a similar conclusion could not be obtained. In the selection of tunnel engineering lines, the axial direction of the tunnel should be adjusted to be consistent with the q_{opt} direction as much as possible to improve the self-stabilizing capability of the surrounding rock.

5.2.2. Effect of the support pressure p_s

The following cases were used to analyze the effect of the support

pressure p_s on the surrounding rock in the plastic zone. The initial geological strength index of the rock mass was $GSI_0 = 60$, residual value was $GSI_r = 26.9$ according to Eq. (8), and p_s values were 0, 1, 2, 3, and 4 MPa. Other calculation parameters were the same as those presented in Section 4. The calculation results are shown in Fig. 13.

Figs. 13 and 14 show the distribution of the circumferential stress σ_θ and radial displacement u_r . The radial stress σ_r and axial stress σ_z had similar variation laws as σ_θ . It can be seen that the stress values at the elastic–plastic interface at different p_s conditions were the same, and the stress curve was approximately obtained by translation. As the support pressure (radial pressure at the cave wall) increased, the value of σ_θ at the tunnel wall increased, and the overall stress level at the cave wall increased. Increasing the value of p_s increased the stress level at the tunnel wall. Consequently, the stress in the plastic zone could increase to the high stress level at the elastic–plastic interface as soon as possible, thereby decreasing the extent of the plastic zone and displacement of the tunnel wall. Increasing the support pressure effectively decreased the

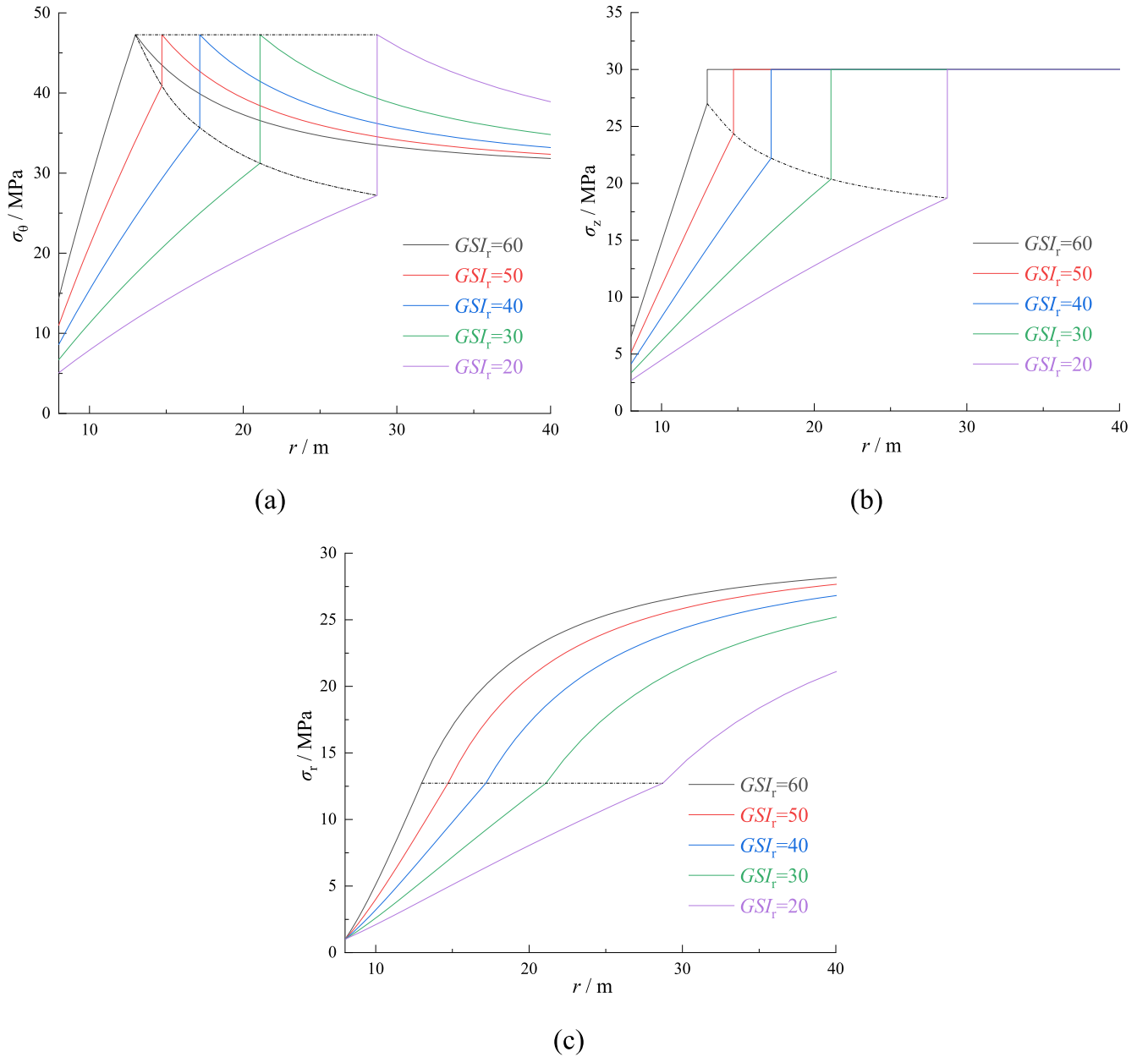


Fig. 15. Stress of the surrounding rock calculated using different values of GSI_r : (a) σ_θ ; (b) σ_z ; and (c) σ_r .

displacement at the tunnel wall; however, when the support pressure was high, the decrease in the deformation of the tunnel wall caused by a further increase in the support pressure yielded minimal benefit.

5.2.3. Effect of the residual strength GSI_r

The mechanical state of the surrounding rock in the plastic zone was closely related to the parameters of the surrounding rock. In this study, the following elastic-brittle model cases were used to analyze the effect of the rock mass parameter GSI_r on the plastic zone. The initial geological strength index of the rock mass was $GSI_0 = 60$. The residual values of GSI_r were 60, 50, 40, 30, and 20. A GSI_r of 60 indicated that a reduction in the rock mass strength parameters was not considered. The elastic-perfect plastic model was used as the control group, and the GZZ strength criterion was used for the calculation. Other calculation parameters were the same as those presented in Section 4.

Fig. 15 shows the variation law of the surrounding rock stress at different GSI_r conditions. For different values of GSI_r , the stress value on the elastic zone side at the elastic-plastic interface was unchanged,

whereas the σ_θ and σ_z on the plastic zone side decreased with a decrease in GSI_r . The smaller the GSI_r , the lower the quality of the surrounding rock in the plastic zone, smaller the stress level, wider the distribution range of the plastic zone, and slower the stress change. It can be seen from σ_θ and σ_r that the smaller the GSI_r , the more obvious the stress drop at the elastic-plastic interface, lower the stress value after the drop, and lower the stress at the tunnel wall, even if the support pressure was the same.

For an insignificant GSI_r , the widely distributed plastic zone yielded significant plastic deformation, and the radial displacement at the tunnel wall increased rapidly, as shown in Fig. 16(a). For instance, at $GSI_r = 20$, $u_{R0} = 1.00$ m, which was 3.2 times that of u_{R0} (0.31 m) at $GSI_r = 60$. Tunnel and unloading redistributed the surrounding rock stress and created a free surface at the tunnel wall. The high-stress difference caused the surrounding rock to be damaged and broken, and the strength decreased. Further, GSI_r was used to reflect the degree of decrease in the rock mass strength. The lower the GSI_r , the higher the degree of rock mass fragmentation, lower the surrounding rock quality,

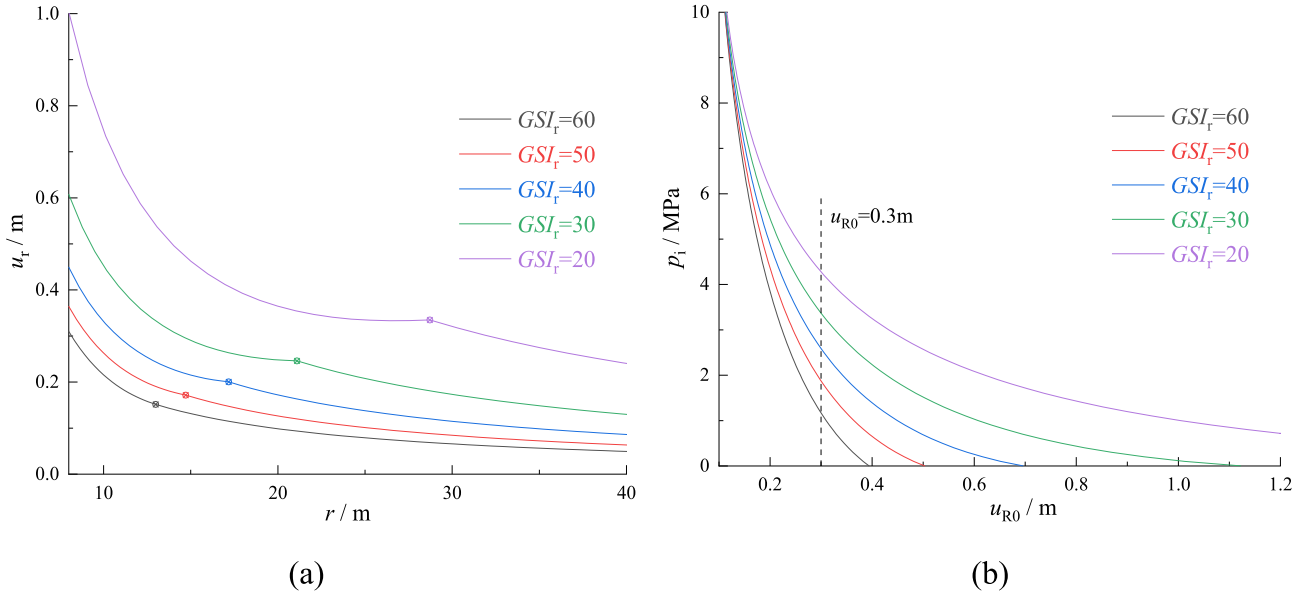


Fig. 16. Results calculated using different values of GSI_r : (a) displacement of tunnel wall; and (b) GRC.

Table 1

Cases and parameters.

	GSI_0	GSI_r	$\psi_0 / ^\circ$	$\psi_r / ^\circ$
Case 1	50	50	15	15
Case 2	50	50	15	0
Case 3	50	40	15	15

and weaker the bearing capacity. Therefore, when the stress level in the plastic zone was insignificant, the plastic zone was significant, and the wall displacement was significant, thus resulting in significant deformation disasters in deep-buried tunnels.

The ground response curves (GRC) in Fig. 16(b) show that when p_s was significant, u_{R0} was insignificant, and the effect of GSI_r could be ignored. With the decrease in p_s , the u_{R0} at different GSI_r conditions distinctly changed, and the difference continued to increase. The required support pressure to control the deformation of the surrounding

rock of the tunnel wall to the same level was several times different. For instance, when the deformation of the surrounding rock was controlled to be 0.3 m, the required support pressure at $GSI_r = 20$ was $p_s = 4.3$ MPa, which was approximately 4 times that at $GSI_r = 60$ ($p_s = 1.1$ MPa). More materials and technological input were needed for support.

5.3. Applicability and limitation

In this study, three sets of cases were used in the calculation and compared using the conventional and proposed methods. The applicability of the proposed method to the strain softening model was verified. The different parameters in the cases are listed in Table 1, and the other parameters are the same as those presented in Section 4. Case 1 was the control group using the elastic-perfect plastic model without considering the change in the strength parameters, or deformation parameters. Case 2 only considered the change in the deformation parameter ψ , and case 3 only considered the softening of the strength parameter GSI . It

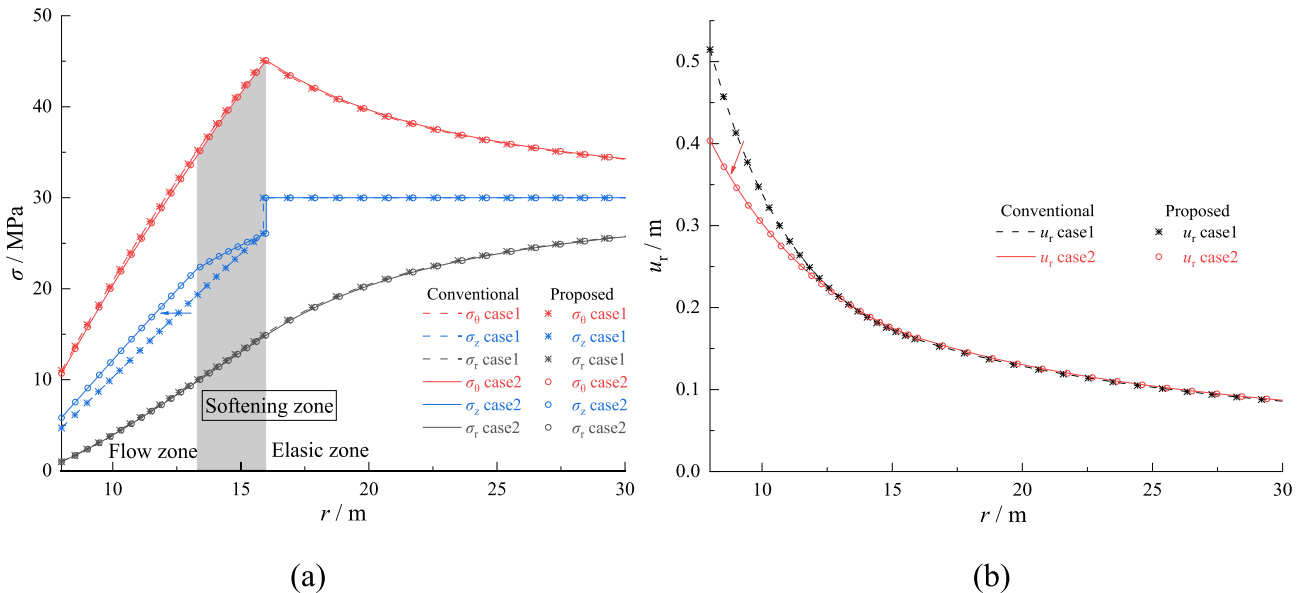


Fig. 17. Results of case 1 and case 2: (a) stress; and (b) tunnel wall displacement.

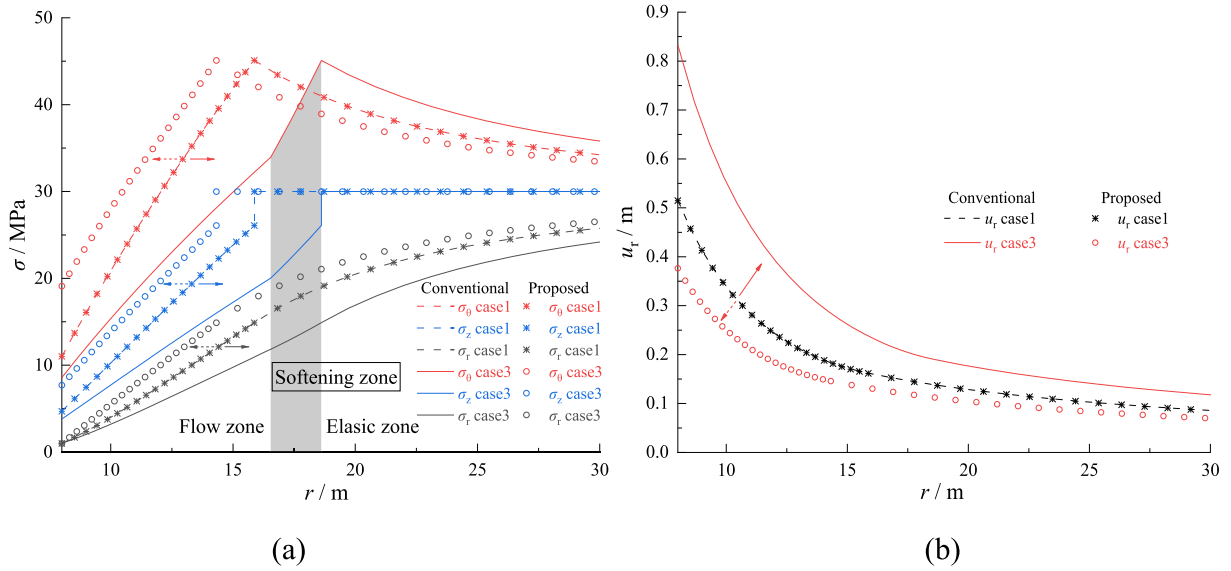


Fig. 18. Results of case 1 and case 3: (a) stress; and (b) tunnel wall displacement.

was assumed that the surrounding rock parameters had a bilinear relationship with the plastic deviatoric strain γ^p , as follows.

$$\omega(\gamma^p) = \begin{cases} \omega_0 - \frac{\omega_0 - \omega_r}{\gamma^{p*}} \gamma^p & (0 < \gamma^p < \gamma^{p*}) \\ \omega_r & (\gamma^p > \gamma^{p*}) \end{cases}, \quad (42)$$

where $\gamma^p = \varepsilon_\theta^p - \varepsilon_r^p$, ε_θ^p and ε_r^p are the circumferential and radial plastic strains, respectively; γ^{p*} is the critical plastic deviatoric strain, such that when γ^p exceeds this value, the surrounding rock parameters are always the residual values, which indicates that the surrounding rock enter the flow zone; ω represents the surrounding rock parameters GSI or ψ ; and ω_0 and ω_r denote the initial and residual values of the parameters, respectively. Herein, $\gamma^{p*} = 0.02$ was used in the calculation cases.

It is observed that σ_z of case 2 changed more considerably than that of case 1 because the σ_z expression (14) was related to the dilation angle, as shown in Fig. 17(a). The slope of the plastic zone σ_z in case 1 was almost unchanged, whereas that in case 2 clearly changed owing to the change in the dilation angle parameter ψ in the softening and flow zones. Therefore, the softening and flow zones could be distinguished according to the different slopes of σ_z . The change in the deformation parameter dilation angle ψ inevitably led to a change in the deformation, as shown in Fig. 17(b). When the dilation angle decreased, the deformation decreased. The change in the dilation angle had a significant effect on σ_z and u_r , but had minimal effect on σ_θ , σ_r and the radius of the plastic zone. It can be seen from the two cases that the calculation results of the two methods were consistent. The proposed method could be applied to the model calculation of the change in the dilation angle of the deformation parameters.

It can be seen from the calculation results of the conventional method (shown by the solid line of case 3 in Fig. 18(a)) that with the softening of strength parameters, the plastic zone of the surrounding rock increased as compared with that of case 1, and the stress curve moved away from the tunnel wall. There was an obvious softening zone. The slopes of the σ_θ and σ_z curves in the softening zone were significantly larger than those in the flow zone. Additionally, the radial displacement increased, as shown in Fig. 18(b). However, the calculation results of the proposed method (shown by the circle points of case 3 in Fig. 18(a)) were opposite those of the conventional method. The stress curve moved toward the cave wall and the slopes of the curves in the softening and flow zones did not change significantly. The radius of the plastic zone and the displacement of the cave wall decreased, which was not

consistent with the engineering practice of the weakening of rock mass strength parameters. Therefore, the calculation results were incorrect; hence, the proposed method was not suitable for the calculation model of strength parameter softening.

The core of the proposed method was the application of Eq. (18). Eqs. (17) and (18) could be established only when the stress was variable and the other strength parameters in the strength criterion were constant. In case 2, the deformation parameter ψ was changed, but the strength parameter was unchanged, and the obtained result was correct; in case 3, the strength parameter GSI was changed and Eq. (18) did not hold, and a completely opposite result was obtained as compared with that of the conventional method. Therefore, the proposed method had acceptable applicability to the models of elastic–perfect plastic, elastic–brittle, and deformation parameter changes. Additionally, it could be used for the excavation analysis of deep-buried rock tunnels with soft surrounding rock and high in situ stress. Its application to the strain–softening model requires further research.

6. Conclusion

(1) This study proposed a circular tunnel excavation analysis method suitable for elastic–brittle and elastic–perfect plastic rock masses. The method was based on a finite difference method. It used linear equations to replace the complex yield surface equation at each point in the plastic zone, avoiding repeated iterative solutions and effectively improving the calculation rate. The case results obtained using the proposed method were correlated with those of the conventional method and 3D finite element calculations. The calculation rate of the stress and strain in the plastic zone using the proposed method was 12,000 times higher than that of the conventional iterative method. The overall calculation rate was 65 times higher, and the calculation error was smaller than those of conventional methods.

(2) The mechanical properties of the surrounding rock were analyzed at different strength criteria (GHB, GPH, and GZZ) using the proposed method. The 2D GHB ignored the effect of the intermediate principal stress, and the calculated deformation of the surrounding rock was larger and the results were more conservative. The GPH could not accurately reflect the strength difference of the rock mass at triaxial tensile and compressive states, and the effect of the intermediate principal stress, thus resulting in significant deviations in the calculation results. The characteristic curves obtained using different criteria showed significant differences. A reasonable 3D strength criterion (e.g.,

GZZ) should be adopted to perform fine analysis of the surrounding rock mechanics and engineering stability design.

(3) An optimal initial axial stress q_{opt} could be obtained using 3D criteria (GPH and GZZ), thus resulting in the smallest calculated results of the tunnel wall displacement u_{R0} and plastic zone radius R_p . Additionally, this resulted in the best stability of the surrounding rock. The q_{opt} obtained using the GZZ was less than the initial stress p , whereas that obtained using the GPH was higher than p . The 2D GHB could not be used to evaluate q_{opt} and conduct engineering design (e.g., tunnel line selection) as it did not consider the effect of the intermediate principal stress, or the axial stress of the surrounding rock. This highlighted the obvious application limitations of the 2D GHB in practical engineering. The support reaction force p_s and the residual strength GSI_r had a significant effect on the convergence–restraint characteristics of the surrounding rock.

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CRediT authorship contribution statement

Chenlong Su: Investigation, Methodology, Software, Data curation, Writing – original draft. **Wuqiang Cai:** Supervision, Conceptualization, Formal analysis, Resources. **Hehua Zhu:** Supervision, Conceptualization, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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None.

Appendix A. Calculation procedure

Input data

Strength parameters: GSI_0 , GSI_r , σ_c , m_i , D .

Deformation parameters: E , ν , ψ .

Model parameters: p , q , p_s , R_0 , n .

At the elastic–plastic interface

On the side of the elastic zone, $i = -1$, $GSI = GSI_0$.

1. Calculate p_{cr} using Eqs. (1) and (11).
2. If $p_s \geq p_{cr}$, there is no plastic zone generated. Calculate the elastic zone solution using.

Eqs. (9) and (10), and end the procedure.

3. Calculate $\sigma_{\theta(-1)}$, $\sigma_{r(-1)}$, $\sigma_{z(-1)}$, $\varepsilon_{\theta(-1)}$, $\varepsilon_{r(-1)}$, $u_{r(-1)}$ using Eq. (11) by setting $r = R_p$.

On the side of the plastic zone, $i = 0$, $GSI = GSI_r$.

4. $\sigma_{r(0)}$, $\varepsilon_{\theta(0)}$, $\varepsilon_{r(0)}$, and $u_{r(0)}$ are the same as $i = -1$.
5. Calculate $\sigma_{\theta(0)}$ and $\sigma_{z(0)}$ using Eqs. (1) and (13).
6. $\varepsilon_{r(0)}^p = 0$, $\varepsilon_{\theta(0)}^p = 0$, $\varepsilon_{r(0)}^e = \varepsilon_{r(0)}$, $\varepsilon_{\theta(0)}^e = \varepsilon_{\theta(0)}$, $\Delta\sigma_r = -(p_{cr} - p_s)/n$, $\rho_{(0)} = 1$.

Plastic zone

for $i = 1$ to n ,

1. $\sigma_{r(i)} = \sigma_{r(0)} + i \cdot \Delta\sigma_r$.
2. Calculate $\partial F / \partial \sigma_{r(i-1)}$, $\partial F / \partial \sigma_{\theta(i-1)}$, $\partial F / \partial \sigma_{z(i-1)}$ using Eq. (B.6) or (B.11) or (B.12).
3. Calculate $c_{\theta 1}$, $c_{z 1}$, $c_{\theta 2}$, $c_{z 2}$, L_1 , L_2 using Eq. (21).
4. $\sigma_{\theta(i)} = (L_1 c_{z 2} - L_2 c_{z 1}) / (c_{\theta 1} c_{z 2} - c_{\theta 2} c_{z 1})$, $\sigma_{z(i)} = (L_1 c_{\theta 2} - L_2 c_{\theta 1}) / (c_{z 1} c_{\theta 2} - c_{z 2} c_{\theta 1})$.
5. $\rho_{(i)} = \rho_{(i-1)} (\sigma_{\theta(i)} - \sigma_{r(i)}) / (\sigma_{\theta(i)} - \sigma_{r(i)} - \Delta\sigma_r)$.
6. $\Delta\varepsilon_{r(i)}^e = [(1 - \nu)\Delta\sigma_r - \nu(\sigma_{\theta(i)} - \sigma_{\theta(i-1)})] / 2G$, $\Delta\varepsilon_{\theta(i)}^e = [(1 - \nu)(\sigma_{\theta(i)} - \sigma_{\theta(i-1)}) - \nu\Delta\sigma_r] / 2G$.
7. $\varepsilon_{r(i)}^e = \varepsilon_{r(i-1)}^e + \Delta\varepsilon_{r(i)}^e$, $\varepsilon_{\theta(i)}^e = \varepsilon_{\theta(i-1)}^e + \Delta\varepsilon_{\theta(i)}^e$.
8. $\lambda = (1 + \sin\psi) / (1 - \sin\psi)$, and $\eta_{(i)} = 1 - r_{(i-1)} / r_{(i)}$.
9. $(\varepsilon_{\theta(i)}^e - \varepsilon_{\theta(i-1)}^e) = [(1 - \nu)(\sigma_{\theta(i)} - \sigma_{\theta(i-1)}) - \nu\Delta\sigma_r] / 2G$, $(\varepsilon_{r(i)}^e - \varepsilon_{r(i-1)}^e) = (\sigma_{\theta(i)} - \sigma_{r(i)}) / 2G$.
10. $\varepsilon_{\theta(i)}^p = -[(\varepsilon_{\theta(i)}^e - \varepsilon_{\theta(i-1)}^e) + \eta_{(i)}(\varepsilon_{\theta(i)}^e - \varepsilon_{r(i)}^e) - \varepsilon_{\theta(i-1)}^p - \eta_{(i)}(\lambda\varepsilon_{\theta(i-1)}^p + \varepsilon_{r(i-1)}^p)] / [1 + (1 + \lambda)\eta_{(i)}]$.
11. $\varepsilon_{r(i)}^p = \varepsilon_{r(i-1)}^p - \lambda(\varepsilon_{\theta(i)}^p - \varepsilon_{r(i-1)}^p)$.
12. $\varepsilon_{r(i)} = \varepsilon_{r(i)}^e + \varepsilon_{r(i)}^p$, $\varepsilon_{\theta(i)} = \varepsilon_{\theta(i)}^e + \varepsilon_{\theta(i)}^p$.

end for

$$13. R_p = R_0/\rho_{(n)}, r = \rho R_p, u_r = \varepsilon_0 r.$$

Finally, calculate the elastic zone solution using Eq. (11).

Appendix B. Calculation coefficient derivation process

The specific expressions of the calculation coefficients $\frac{\partial F}{\partial \sigma_{k(i-1)}}$ ($k = r, \theta, z$) for different strength criteria are given below.

Taking the GZZ strength criterion as an example, the partial derivative of the yield function with respect to the stress tensor σ can be written as:

$$\frac{\partial F_{GZZ}}{\partial \sigma} = \frac{\partial F_{GZZ}}{\partial I_1} \frac{\partial I_1}{\partial \sigma} + \frac{\partial F_{GZZ}}{\partial \sqrt{J_2}} \frac{\partial \sqrt{J_2}}{\partial \sigma} + \frac{\partial F_{GZZ}}{\partial \theta_\sigma} \frac{\partial \theta_\sigma}{\partial \sigma}, \quad (B.1)$$

where the partial derivative of the strength criterion to each stress invariant is:

$$\begin{cases} \frac{\partial F_{GZZ}}{\partial I_1} = -\frac{1}{3}m_b \\ \frac{\partial F_{GZZ}}{\partial \sqrt{J_2}} = \frac{\sqrt{3}}{a\sigma_c^{1/a-1}} \left(\sqrt{3}J_2 \right)^{1/a-1} + \left(\frac{3 + \sin 3\theta_\sigma}{2\sqrt{3}} \right) m_b \\ \frac{\partial F_{GZZ}}{\partial \theta_\sigma} = \frac{\sqrt{3} m_b \cos 3\theta_\sigma}{2} \sqrt{J_2} \end{cases} \quad (B.2)$$

Deriving Eq. (7) of the relationship between θ_σ and J_2, J_3 with respect to the stress tensor σ , the simplification can be obtained using Eq. (7):

$$\frac{\partial \theta_\sigma}{\partial \sigma} = -\frac{\tan 3\theta_\sigma}{\sqrt{J_2}} \frac{\partial \sqrt{J_2}}{\partial \sigma} - \frac{\sqrt{3}}{2\sqrt{J_2^3 \cos 3\theta_\sigma}} \frac{\partial J_3}{\partial \sigma}. \quad (B.3)$$

The partial derivative of stress invariant to stress tensor is:

$$\begin{cases} \frac{\partial I_1}{\partial \sigma} = \mathbf{I} \\ \frac{\partial \sqrt{J_2}}{\partial \sigma} = \frac{\mathbf{S}}{2\sqrt{J_2}} \\ \frac{\partial J_3}{\partial \sigma} = \mathbf{S} \cdot \mathbf{S} - \frac{2J_2}{3} \mathbf{I} \end{cases} \quad (B.4)$$

where $\mathbf{I}$ is the unit matrix and $\mathbf{S}$ is the stress partial tensor. The following equation can be obtained by substituting Eqs. (B.2), (B.3), and (B.4) into Eq. (B.1):

$$\frac{\partial F_{GZZ}}{\partial \sigma} = m_b \left[\frac{1}{6} + \frac{\xi(J_2, \theta_\sigma)}{2\sqrt{J_2}} \mathbf{S} - \frac{3}{4J_2} \mathbf{S} \cdot \mathbf{S} \right], \quad (B.5)$$

$$\text{where } \xi(J_2, \theta_\sigma) = \sqrt{3} \left[\frac{(\sqrt{3}J_2)^{1/a-1}}{a m_b \sigma_c^{1/a-1}} - \frac{\sin 3\theta_\sigma}{3} + \frac{1}{2} \right].$$

Thus, the calculation coefficient of the GZZ strength criterion is:

$$\begin{bmatrix} \frac{\partial F_{GZZ}}{\partial \sigma_{r(i-1)}} \\ \frac{\partial F_{GZZ}}{\partial \sigma_{\theta(i-1)}} \\ \frac{\partial F_{GZZ}}{\partial \sigma_{z(i-1)}} \end{bmatrix} = \frac{m_b}{2} \begin{bmatrix} S_{r(i-1)} & S_{r(i-1)}^2 & 1 \\ S_{\theta(i-1)} & S_{\theta(i-1)}^2 & 1 \\ S_{z(i-1)} & S_{z(i-1)}^2 & 1 \end{bmatrix} \begin{bmatrix} \frac{\xi(J_{2(i-1)}, \theta_{\sigma(i-1)})}{\sqrt{J_{2(i-1)}}} \\ -\frac{3}{2J_{2(i-1)}} \\ \frac{1}{3} \end{bmatrix}. \quad (B.6)$$

Similarly, the partial derivative of the GPH yield function to σ can be written as:

$$\frac{\partial F_{GPH}}{\partial \sigma} = \frac{\partial F_{GPH}}{\partial I_1} \frac{\partial I_1}{\partial \sigma} + \frac{\partial F_{GPH}}{\partial \tau_{oct}} \frac{\partial \tau_{oct}}{\partial \sigma}. \quad (B.7)$$

The partial derivative of the yield function to each stress invariant is:

$$\begin{cases} \frac{\partial F_{GPH}}{\partial I_1} = -\frac{1}{3}m_b \\ \frac{\partial F_{GPH}}{\partial \tau_{oct}} = \frac{3}{\sqrt{2} a \sigma_c^{1/a-1}} \left(\frac{3\tau_{oct}}{\sqrt{2}} \right)^{1/a-1} + \frac{3}{2\sqrt{2}} m_b \end{cases} \quad (B.8)$$

The partial derivative of stress invariant to stress tensor is:

$$\begin{cases} \frac{\partial I_1}{\partial \boldsymbol{\sigma}} = \mathbf{I} \\ \frac{\partial \tau_{\text{oct}}}{\partial \boldsymbol{\sigma}} = \frac{\mathbf{S}}{3\tau_{\text{oct}}} \end{cases} \quad (\text{B.9})$$

Substituting Eqs. (B.8) and (B.9) into Eq. (B.7) to get:

$$\frac{\partial F_{\text{GPH}}}{\partial \boldsymbol{\sigma}} = -\frac{1}{3}m_b \mathbf{I} + \left(\frac{1}{\sqrt{2}a\sigma_c^{1/a-1}} \left(\frac{3\tau_{\text{oct}}}{\sqrt{2}} \right)^{1/a-1} + \frac{1}{2\sqrt{2}}m_b \right) \frac{\mathbf{S}}{\tau_{\text{oct}}} \quad (\text{B.10})$$

Thus, the calculation coefficient of the GPH strength criterion is:

$$\begin{bmatrix} \frac{\partial F_{\text{GPH}}}{\partial \sigma_{r(i-1)}} \\ \frac{\partial F_{\text{GPH}}}{\partial \sigma_{\theta(i-1)}} \\ \frac{\partial F_{\text{GPH}}}{\partial \sigma_{z(i-1)}} \end{bmatrix} = -\frac{1}{3}m_b + \frac{1}{\tau_{\text{oct}}} \left(\frac{1}{\sqrt{2}a\sigma_c^{1/a-1}} \left(\frac{3\tau_{\text{oct}}}{\sqrt{2}} \right)^{1/a-1} + \frac{1}{2\sqrt{2}}m_b \right) \begin{bmatrix} S_{r(i-1)} \\ S_{\theta(i-1)} \\ S_{z(i-1)} \end{bmatrix} \quad (\text{B.11})$$

The form of the GHB is simple. The yield function can directly use the derivative of $\boldsymbol{\sigma}$. The calculation coefficient of the GHB strength criterion is:

$$\begin{bmatrix} \frac{\partial F_{\text{GHB}}}{\partial \sigma_{r(i-1)}} \\ \frac{\partial F_{\text{GHB}}}{\partial \sigma_{\theta(i-1)}} \\ \frac{\partial F_{\text{GHB}}}{\partial \sigma_{z(i-1)}} \end{bmatrix} = \begin{bmatrix} -\frac{1}{a\sigma_c^{1/a-1}} (\sigma_{\theta(i-1)} - \sigma_{r(i-1)})^{1/a-1} - m_b \\ \frac{1}{a\sigma_c^{1/a-1}} (\sigma_{\theta(i-1)} - \sigma_{r(i-1)})^{1/a-1} \\ 0 \end{bmatrix} \quad (\text{B.12})$$

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